SKALA FULL-STACK ENGINEERING · 종합실습 4

E-Commerce 매출 분석 및 정리

랜덤 데이터 위에서 요청을 분석 조건으로 바꾸고, 검증 가능한 결과로 전달하기

판교 4반 · 최승우

PostgreSQL 17.10 · ecom schema

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보고서 관점 이번 실습은 가장 빠른 SQL을 고르는 시험이 아니라, 구매·재무·CRM의 요청을 같은 기준으로 재현하고 설명하는 연습이다.

Executive Summary

분석의 핵심은 '누가 요청했는가'보다 '무엇을 한 건으로 볼 것인가'를 먼저 정하는 것이었다. 주문과 주문상품의 1:N 관계를 그대로 집계하면 주문 수와 평균 주문 금액이 왜곡되므로, Q2·Q5·Q9는 주문 단위 선집계를 기준으로 재작성했다. 상관 서브쿼리가 반복되던 Q6·Q7·Q10은 윈도 함수·JOIN·조건부 집계로 단순화했다.

분석 DB	skala_ecom_practice4 / user=chltmddn5843
데이터 규모	고객 3,000 · 상품 600 · 주문 9,890 · 주문상품 26,716
유효 매출 상태	paid · shipped · delivered
공통 매출식	order_items.line_total = unit_price × qty - discount
측정 방식	비교표는 EXPLAIN 3회 중앙값 · 원문 로그 화면은 실행 당시 1회 실제 출력

권한 주의 본 실습은 로컬 관리자 계정으로 수행했다. 운영 환경에서는 분석용 읽기 전용 계정과 Materialized View 유지보수 계정을 분리하고 최소 권한만 부여해야 한다.

요청 대응 원칙

- 기간·상태·집계 단위·출력 건수를 SQL보다 먼저 확정한다.
- 랜덤 데이터에서는 수치보다 결과가 바뀌어도 유지되는 판정 로직을 검증한다.
- 실행시간만 보지 않고 실제 행 수와 Buffers를 함께 비교한다.
- 결과에는 기준시각과 가정을 붙이고, 관찰 비교를 인과효과로 표현하지 않는다.

Q1. 최근 한 달 실제 총매출

요청 부서	재무
요청 목적	최근 한 달 실제 매출 규모 확인
판정 기준	롤링 1개월·유효 주문 상태

튜닝 전

직접 조인 집계

```
SELECT
            sum(oi.line_total) AS total_revenue
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
          AND o.order_ts >= now() - interval '1 month'
```

튜닝 전 실행 결과

Q01 튜닝 전 결과

```
total_revenue
-------------
2,058,131.00
rows=1 | median execution=5.237 ms
```

튜닝 전 성능 원문 로그 · Q01_plan_before.png

```
Q01 최근 한 달 실제 총매출 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
        sum(oi.line_total) AS total_revenue
      FROM ecom.orders o
      JOIN ecom.order_items oi
        ON oi.order_id = o.order_id
      WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        AND o.order_ts >= now() - interval '1 month'

PERFORMANCE RESULT
Aggregate  (cost=947.69..947.70 rows=1 width=32) (actual time=8.418..8.422 rows=1 loops=1)
  Buffers: shared hit=360
  ->  Hash Join  (cost=338.26..933.59 rows=5640 width=6) (actual time=3.514..7.968 rows=5756 loops=1)
        Hash Cond: (oi.order_id = o.order_id)
        Buffers: shared hit=360
        ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual time=0.005..1.752
rows=26716 loops=1)
              Buffers: shared hit=258
        ->  Hash  (cost=312.16..312.16 rows=2088 width=8) (actual time=3.489..3.490 rows=2111 loops=1)
              Buckets: 4096  Batches: 1  Memory Usage: 115kB
              Buffers: shared hit=102
              ->  Seq Scan on orders o  (cost=0.00..312.16 rows=2088 width=8) (actual time=0.010..3.245
rows=2111 loops=1)
                    Filter: ((order_status = ANY ('{paid,shipped,delivered}'::text[])) AND (order_ts >=
(now() - '1 mon'::interval)))
                    Rows Removed by Filter: 7779
                    Buffers: shared hit=102
Planning:
  Buffers: shared hit=357
Planning Time: 1.299 ms
Execution Time: 8.473 ms
```

튜닝 포인트

변경 기간·상태 대상 주문을 먼저 명명해 조건과 집계 경로를 분리

튜닝 후

```
WITH recent_orders AS (
            SELECT
                order_id
            FROM ecom.orders
            WHERE order_status IN ('paid', 'shipped', 'delivered')
              AND order_ts >= now() - interval '1 month'
        )

        SELECT
            sum(oi.line_total) AS total_revenue
        FROM recent_orders ro
        JOIN ecom.order_items oi
          ON oi.order_id = ro.order_id
```

튜닝 후 실행 결과

Q01 튜닝 후 결과

```
total_revenue
-------------
2,058,131.00
rows=1 | median execution=4.364 ms
```

튜닝 후 성능 원문 로그 · Q01_plan_after.png

```
Q01 최근 한 달 실제 총매출 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH recent_orders AS (
          SELECT
             order_id
          FROM ecom.orders
          WHERE order_status IN ('paid', 'shipped', 'delivered')
            AND order_ts >= now() - interval '1 month'
       )

       SELECT
          sum(oi.line_total) AS total_revenue
       FROM recent_orders ro
       JOIN ecom.order_items oi
         ON oi.order_id = ro.order_id

PERFORMANCE RESULT
Aggregate  (cost=947.69..947.70 rows=1 width=32) (actual time=8.299..8.303 rows=1 loops=1)
  Buffers: shared hit=360
  ->  Hash Join  (cost=338.26..933.59 rows=5640 width=6) (actual time=3.102..7.814 rows=5756 loops=1)
        Hash Cond: (oi.order_id = orders.order_id)
        Buffers: shared hit=360
        ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual time=0.004..1.892
rows=26716 loops=1)
              Buffers: shared hit=258
        ->  Hash  (cost=312.16..312.16 rows=2088 width=8) (actual time=3.078..3.078 rows=2111 loops=1)
              Buckets: 4096  Batches: 1  Memory Usage: 115kB
              Buffers: shared hit=102
              ->  Seq Scan on orders  (cost=0.00..312.16 rows=2088 width=8) (actual time=0.009..2.857
rows=2111 loops=1)
                    Filter: ((order_status = ANY ('{paid,shipped,delivered}'::text[])) AND (order_ts >=
(now() - '1 mon'::interval)))
                    Rows Removed by Filter: 7779
                    Buffers: shared hit=102
Planning:
  Buffers: shared hit=357
Planning Time: 1.156 ms
Execution Time: 8.346 ms
```

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	5.237 ms	4.364 ms
Planning Time	0.094 ms	0.056 ms
Actual Rows	1	1
Shared Hit	360	360
Shared Read	0	0
Top Plan Node	Aggregate	Aggregate
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

PostgreSQL이 CTE를 인라인하면 계획은 같을 수 있다. 성능보다 요청 조건의 가독성이 핵심이다.

재무 전달안

최근 한 달 실제 매출은 2,058,131.00이다.

Q2. 월별 주문 수·매출·AOV

요청 부서	재무·경영
요청 목적	월별 주문 수·매출·AOV 추이
판정 기준	달력 월·주문 단위 평균

튜닝 전

주문상품 행에서 COUNT DISTINCT와 합계를 동시에 수행

```
SELECT
            date_trunc('month', o.order_ts)::date AS month,
            count(DISTINCT o.order_id) AS order_count,
            sum(oi.line_total) AS revenue,
            sum(oi.line_total) / count(DISTINCT o.order_id) AS aov
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY date_trunc('month', o.order_ts)::date
        ORDER BY month
```

튜닝 전 실행 결과

Q02 튜닝 전 결과

```
month      | order_count | revenue      | aov
-----------+-------------+--------------+--------
2026-04-01 | 823         | 728,085.26   | 884.67
2026-05-01 | 1873        | 1,666,271.77 | 889.63
2026-06-01 | 1915        | 1,687,798.91 | 881.36
2026-07-01 | 2144        | 2,034,521.03 | 948.94
2026-08-01 | 858         | 871,025.70   | 1,015.18
rows=5 | median execution=10.604 ms
```

튜닝 전 성능 원문 로그 · Q02_plan_before.png

```
Q02 월별 주문 수·매출·AOV [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
          date_trunc('month', o.order_ts)::date AS month,
          count(DISTINCT o.order_id) AS order_count,
          sum(oi.line_total) AS revenue,
          sum(oi.line_total) / count(DISTINCT o.order_id) AS aov
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY date_trunc('month', o.order_ts)::date
        ORDER BY month

PERFORMANCE RESULT
GroupAggregate  (cost=2504.56..2881.51 rows=7613 width=76) (actual time=14.644..16.750 rows=5 loops=1)
  Group Key: ((date_trunc('month'::text, o.order_ts))::date)
  Buffers: shared hit=366
  ->  Sort  (cost=2504.56..2555.98 rows=20565 width=18) (actual time=14.462..15.102 rows=20557 loops=1)
        Sort Key: ((date_trunc('month'::text, o.order_ts))::date), o.order_id
        Sort Method: quicksort  Memory: 1572kB
        Buffers: shared hit=366
        ->  Hash Join  (cost=333.15..1031.30 rows=20565 width=18) (actual time=1.588..10.974 rows=20557
loops=1)
              Hash Cond: (oi.order_id = o.order_id)
              Buffers: shared hit=360
              ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.003..1.413 rows=26716 loops=1)
                    Buffers: shared hit=258
              ->  Hash  (cost=237.99..237.99 rows=7613 width=16) (actual time=1.566..1.566 rows=7613
loops=1)
                    Buckets: 8192  Batches: 1  Memory Usage: 421kB
                    Buffers: shared hit=102
                    ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=16) (actual
time=0.004..1.116 rows=7613 loops=1)
                          Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                          Rows Removed by Filter: 2277
                          Buffers: shared hit=102
Planning:
  Buffers: shared hit=280
Planning Time: 0.515 ms
Execution Time: 16.797 ms
```

튜닝 포인트

변경 주문별 금액 선집계 후 월 집계

튜닝 후

```
WITH order_totals AS (
            SELECT
                o.order_id,
                date_trunc('month', o.order_ts)::date AS month,
                sum(oi.line_total) AS amount
            FROM ecom.orders o
            JOIN ecom.order_items oi
              ON oi.order_id = o.order_id
            WHERE o.order_status IN ('paid', 'shipped', 'delivered')
            GROUP BY
                o.order_id,
                date_trunc('month', o.order_ts)::date
        )

        SELECT
            month,
            count(*) AS order_count,
            sum(amount) AS revenue,
            ecom.safe_div(sum(amount), count(*)) AS aov
        FROM order_totals
        GROUP BY month
        ORDER BY month
```

튜닝 후 실행 결과

Q02 튜닝 후 결과

```
month      | order_count | revenue      | aov
-----------+-------------+--------------+---------
2026-04-01 | 823         | 728,085.26   | 884.67
2026-05-01 | 1873        | 1,666,271.77 | 889.63
2026-06-01 | 1915        | 1,687,798.91 | 881.36
2026-07-01 | 2144        | 2,034,521.03 | 948.94
2026-08-01 | 858         | 871,025.70   | 1,015.18
rows=5 | median execution=7.755 ms
```

튜닝 후 성능 원문 로그 · Q02_plan_after.png (1/2)

```
Q02 월별 주문 수·매출·AOV [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH order_totals AS (
        SELECT
            o.order_id,
            date_trunc('month', o.order_ts)::date AS month,
            sum(oi.line_total) AS amount
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY
            o.order_id,
            date_trunc('month', o.order_ts)::date
    )

    SELECT
        month,
        count(*) AS order_count,
        sum(amount) AS revenue,
        ecom.safe_div(sum(amount), count(*)) AS aov
    FROM order_totals
    GROUP BY month
    ORDER BY month

PERFORMANCE RESULT
Sort  (cost=1464.63..1465.13 rows=200 width=76) (actual time=13.735..13.737 rows=5 loops=1)
  Sort Key: ((date_trunc('month'::text, o.order_ts))::date)
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=363
  ->  HashAggregate  (cost=1451.99..1456.99 rows=200 width=76) (actual time=13.726..13.728 rows=5
loops=1)
        Group Key: ((date_trunc('month'::text, o.order_ts))::date)
        Batches: 1  Memory Usage: 40kB
        Buffers: shared hit=360
        ->  HashAggregate  (cost=1185.54..1318.76 rows=7613 width=44) (actual time=12.010..13.063
rows=7613 loops=1)
              Group Key: o.order_id, (date_trunc('month'::text, o.order_ts))::date
              Batches: 1  Memory Usage: 3217kB
              Buffers: shared hit=360
              ->  Hash Join  (cost=333.15..1031.30 rows=20565 width=18) (actual time=1.071..9.196
rows=20557 loops=1)
                    Hash Cond: (oi.order_id = o.order_id)
                    Buffers: shared hit=360
                    ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.002..1.188 rows=26716 loops=1)
                          Buffers: shared hit=258
                    ->  Hash  (cost=237.99..237.99 rows=7613 width=16) (actual time=1.057..1.057
rows=7613 loops=1)
                          Buckets: 8192  Batches: 1  Memory Usage: 421kB
                          Buffers: shared hit=102
                          ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=16) (actual
time=0.002..0.732 rows=7613 loops=1)
                                Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                Rows Removed by Filter: 2277
                                Buffers: shared hit=102
Planning:
  Buffers: shared hit=498
Planning Time: 0.507 ms
Execution Time: 13.810 ms
```

튜닝 후 성능 원문 로그 · Q02_plan_after.png (2/2)

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	10.604 ms	7.755 ms
Planning Time	0.059 ms	0.055 ms
Actual Rows	5	5
Shared Hit	360	360
Shared Read	0	0
Top Plan Node	Aggregate	Sort
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

1:N 조인으로 부푼 행을 주문 단위로 먼저 축소했다.

재무·경영 전달안

최신 월(2026-08-01) 주문은 858건, 매출 871,025.70, AOV 1,015.18로 집계됐다. 부분 월 여부를 함께 전달해야 한다.

Q3. 최근 90일 카테고리 매출 Top 10

요청 부서	구매·MD
요청 목적	최근 90일 카테고리 성과 비교
판정 기준	카테고리 매출 Top 10

튜닝 전

전체 조인 후 기간 필터와 집계

```
SELECT
            c.category_id,
            c.category_name,
            sum(oi.line_total) AS revenue
        FROM ecom.order_items oi
        JOIN ecom.orders o
          ON o.order_id = oi.order_id
        JOIN ecom.products p
          ON p.product_id = oi.product_id
        JOIN ecom.categories c
          ON c.category_id = p.category_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
          AND o.order_ts >= now() - interval '90 days'
        GROUP BY
            c.category_id,
            c.category_name
        ORDER BY
            revenue DESC,
            c.category_id
        LIMIT 10
```

튜닝 전 실행 결과

Q03 튜닝 전 결과

```
category_id | category_name | revenue
------------+---------------+-------------
6           | Appliances    | 1,900,591.52
4           | Audio         | 1,255,708.21
13          | Outdoor       | 309,879.86
3           | Laptops       | 306,953.82
14          | Fitness       | 303,426.89
11          | Shoes         | 286,309.97
7           | Cookware      | 277,215.27
10          | Women         | 270,003.37
... 2 rows omitted
rows=10 | median execution=5.819 ms
```

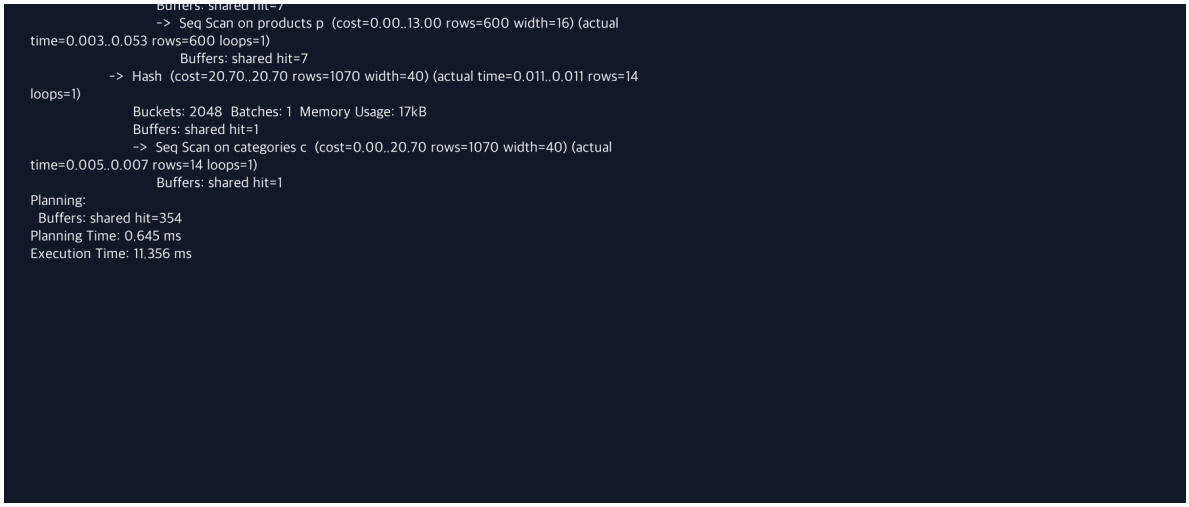
튜닝 전 성능 원문 로그 · Q03_plan_before.png (1/2)

```
Q03 최근 90일 카테고리 매출 Top 10 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
          c.category_id,
          c.category_name,
          sum(oi.line_total) AS revenue
        FROM ecom.order_items oi
        JOIN ecom.orders o
          ON o.order_id = oi.order_id
        JOIN ecom.products p
          ON p.product_id = oi.product_id
        JOIN ecom.categories c
          ON c.category_id = p.category_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
          AND o.order_ts >= now() - interval '90 days'
        GROUP BY
          c.category_id,
          c.category_name
        ORDER BY
          revenue DESC,
          c.category_id
        LIMIT 10

PERFORMANCE RESULT
Limit  (cost=1233.23..1233.25 rows=10 width=72) (actual time=11.279..11.283 rows=10 loops=1)
  Buffers: shared hit=374
  ->  Sort  (cost=1233.23..1235.90 rows=1070 width=72) (actual time=11.278..11.280 rows=10 loops=1)
        Sort Key: (sum(oi.line_total)) DESC, c.category_id
        Sort Method: quicksort  Memory: 25kB
        Buffers: shared hit=374
        ->  HashAggregate  (cost=1196.73..1210.11 rows=1070 width=72) (actual time=11.249..11.259 rows=10
loops=1)
              Group Key: c.category_id
              Batches: 1  Memory Usage: 73kB
              Buffers: shared hit=368
              ->  Hash Join  (cost=439.59..1118.02 rows=15743 width=46) (actual time=3.496..9.888
rows=15883 loops=1)
                    Hash Cond: (p.category_id = c.category_id)
                    Buffers: shared hit=368
                    ->  Hash Join  (cost=405.51..1042.45 rows=15743 width=14) (actual time=3.481..8.406
rows=15883 loops=1)
                          Hash Cond: (oi.product_id = p.product_id)
                          Buffers: shared hit=367
                          ->  Hash Join  (cost=385.01..980.34 rows=15743 width=14) (actual
time=3.379..6.821 rows=15883 loops=1)
                                Hash Cond: (oi.order_id = o.order_id)
                                Buffers: shared hit=360
                                ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=22)
(actual time=0.002..1.117 rows=26716 loops=1)
                                      Buffers: shared hit=258
                                ->  Hash  (cost=312.16..312.16 rows=5828 width=8) (actual
time=3.367..3.368 rows=5873 loops=1)
                                      Buckets: 8192  Batches: 1  Memory Usage: 294kB
                                      Buffers: shared hit=102
                                      ->  Seq Scan on orders o  (cost=0.00..312.16 rows=5828 width=8)
(actual time=0.005..2.962 rows=5873 loops=1)
                                            Filter: ((order_status = ANY
('{paid,shipped,delivered}'::text[])) AND (order_ts >= (now() - '90 days'::interval)))
                                            Rows Removed by Filter: 4017
                                            Buffers: shared hit=102
                          ->  Hash  (cost=13.00..13.00 rows=600 width=16) (actual time=0.097..0.097
rows=600 loops=1)
                                Buckets: 1024  Batches: 1  Memory Usage: 37kB
                                Buffers: shared hit=7
```

튜닝 전 성능 원문 로그 · Q03_plan_before.png (2/2)



```
                        Buffers: shared hit=7
                        ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=16) (actual
time=0.003..0.053 rows=600 loops=1)
                              Buffers: shared hit=7
                  ->  Hash  (cost=20.70..20.70 rows=1070 width=40) (actual time=0.011..0.011 rows=14
loops=1)
                        Buckets: 2048  Batches: 1  Memory Usage: 17kB
                        Buffers: shared hit=1
                        ->  Seq Scan on categories c  (cost=0.00..20.70 rows=1070 width=40) (actual
time=0.005..0.007 rows=14 loops=1)
                              Buffers: shared hit=1
Planning:
  Buffers: shared hit=354
Planning Time: 0.645 ms
Execution Time: 11.356 ms
```

튜닝 포인트

변경 최근 판매행을 먼저 좁힌 뒤 상품·카테고리 연결

튜닝 후

```
WITH recent_sales AS (
            SELECT
                oi.product_id,
                oi.line_total
            FROM ecom.orders o
            JOIN ecom.order_items oi
              ON oi.order_id = o.order_id
            WHERE o.order_status IN ('paid', 'shipped', 'delivered')
              AND o.order_ts >= now() - interval '90 days'
        )

        SELECT
            c.category_id,
            c.category_name,
            sum(rs.line_total) AS revenue
        FROM recent_sales rs
        JOIN ecom.products p
          ON p.product_id = rs.product_id
        JOIN ecom.categories c
          ON c.category_id = p.category_id
        GROUP BY
            c.category_id,
            c.category_name
        ORDER BY
            revenue DESC,
            c.category_id
        LIMIT 10
```

튜닝 후 실행 결과

Q03 튜닝 후 결과

```
category_id | category_name | revenue
------------+---------------+------------
6           | Appliances    | 1,900,591.52
4           | Audio         | 1,255,708.21
13          | Outdoor       | 309,879.86
3           | Laptops       | 306,953.82
14          | Fitness       | 303,426.89
11          | Shoes         | 286,309.97
7           | Cookware      | 277,215.27
10          | Women         | 270,003.37
... 2 rows omitted
rows=10 | median execution=5.760 ms
```

튜닝 후 성능 원문 로그 · Q03_plan_after.png (1/2)

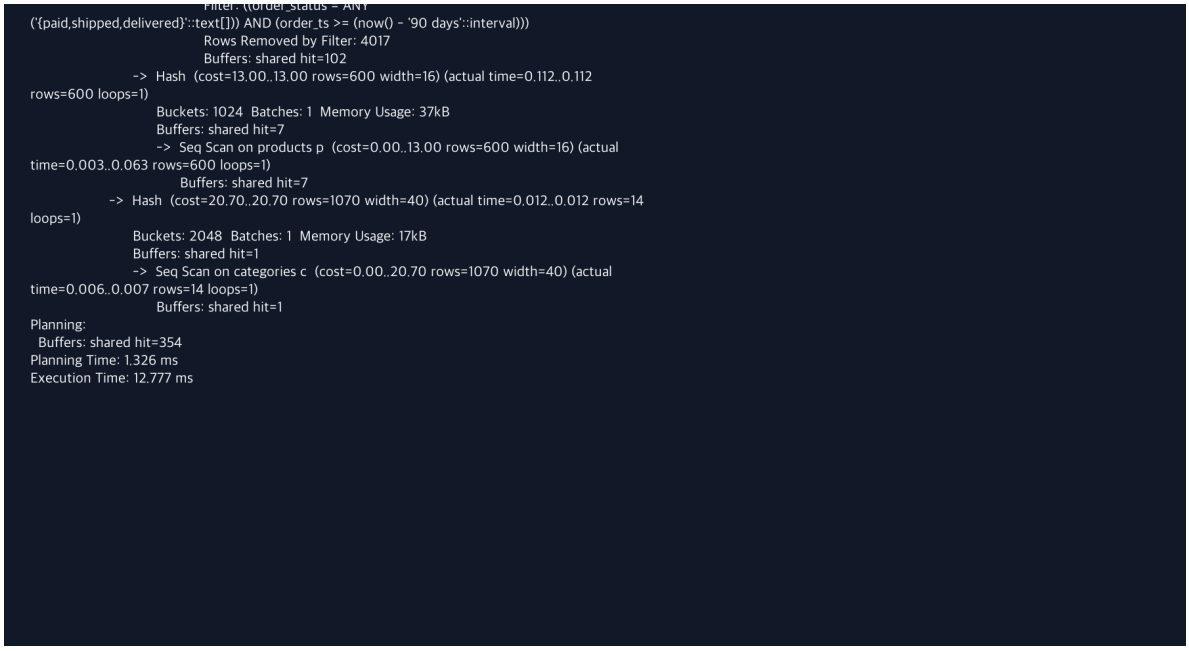
```
Q03 최근 90일 카테고리 매출 Top 10 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH recent_sales AS (
        SELECT
          oi.product_id,
          oi.line_total
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
          AND o.order_ts >= now() - interval '90 days'
    )

    SELECT
        c.category_id,
        c.category_name,
        sum(rs.line_total) AS revenue
    FROM recent_sales rs
    JOIN ecom.products p
      ON p.product_id = rs.product_id
    JOIN ecom.categories c
      ON c.category_id = p.category_id
    GROUP BY
        c.category_id,
        c.category_name
    ORDER BY
        revenue DESC,
        c.category_id
    LIMIT 10

PERFORMANCE RESULT
Limit  (cost=1233.23..1233.25 rows=10 width=72) (actual time=12.705..12.710 rows=10 loops=1)
  Buffers: shared hit=374
  ->  Sort  (cost=1233.23..1235.90 rows=1070 width=72) (actual time=12.704..12.706 rows=10 loops=1)
        Sort Key: (sum(oi.line_total)) DESC, c.category_id
        Sort Method: quicksort  Memory: 25kB
        Buffers: shared hit=374
        ->  HashAggregate  (cost=1196.73..1210.11 rows=1070 width=72) (actual time=12.671..12.683 rows=10
loops=1)
              Group Key: c.category_id
              Batches: 1  Memory Usage: 73kB
              Buffers: shared hit=368
              ->  Hash Join  (cost=439.59..1118.02 rows=15743 width=46) (actual time=3.649..11.126
rows=15883 loops=1)
                    Hash Cond: (p.category_id = c.category_id)
                    Buffers: shared hit=368
                    ->  Hash Join  (cost=405.51..1042.45 rows=15743 width=14) (actual time=3.632..9.422
rows=15883 loops=1)
                          Hash Cond: (oi.product_id = p.product_id)
                          Buffers: shared hit=367
                          ->  Hash Join  (cost=385.01..980.34 rows=15743 width=14) (actual
time=3.513..7.633 rows=15883 loops=1)
                                Hash Cond: (oi.order_id = o.order_id)
                                Buffers: shared hit=360
                                ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=22)
(actual time=0.003..1.382 rows=26716 loops=1)
                                      Buffers: shared hit=258
                                ->  Hash  (cost=312.16..312.16 rows=5828 width=8) (actual
time=3.494..3.495 rows=5873 loops=1)
                                      Buckets: 8192  Batches: 1  Memory Usage: 294kB
                                      Buffers: shared hit=102
                                      ->  Seq Scan on orders o  (cost=0.00..312.16 rows=5828 width=8)
(actual time=0.005..2.987 rows=5873 loops=1)
                                            Filter: ((order_status = ANY
```

튜닝 후 성능 원문 로그 · Q03_plan_after.png (2/2)



```
                                    Filter: ((order_status = ANY
('{paid,shipped,delivered}'::text[])) AND (order_ts >= (now() - '90 days'::interval)))
                                    Rows Removed by Filter: 4017
                                    Buffers: shared hit=102
                        ->  Hash  (cost=13.00..13.00 rows=600 width=16) (actual time=0.112..0.112
rows=600 loops=1)
                              Buckets: 1024  Batches: 1  Memory Usage: 37kB
                              Buffers: shared hit=7
                              ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=16) (actual
time=0.003..0.063 rows=600 loops=1)
                                    Buffers: shared hit=7
                  ->  Hash  (cost=20.70..20.70 rows=1070 width=40) (actual time=0.012..0.012 rows=14
loops=1)
                        Buckets: 2048  Batches: 1  Memory Usage: 17kB
                        Buffers: shared hit=1
                        ->  Seq Scan on categories c  (cost=0.00..20.70 rows=1070 width=40) (actual
time=0.006..0.007 rows=14 loops=1)
                              Buffers: shared hit=1
Planning:
  Buffers: shared hit=354
Planning Time: 1.326 ms
Execution Time: 12.777 ms
```

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	5.819 ms	5.760 ms
Planning Time	0.116 ms	0.153 ms
Actual Rows	10	10
Shared Hit	368	368
Shared Read	0	0
Top Plan Node	Limit	Limit
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

소규모 데이터에서는 계획이 같을 수 있으나 분석 경계가 선명해졌다.

구매·MD 전달안

최근 90일 1위 카테고리는 Appliances, 매출은 1,900,591.52다. 순위는 랜덤 시드 재적재 시 바뀔 수 있다.

Q4. 상품별 누적매출 RANK Top 20

요청 부서	MD
요청 목적	누적매출 상위 상품 선별
판정 기준	RANK 기준 정확히 20개

튜닝 전

상품명까지 포함해 집계와 순위를 한 단계에서 처리

```
SELECT
            p.product_id,
            p.product_name,
            sum(oi.line_total) AS revenue,
            rank() OVER (
                ORDER BY sum(oi.line_total) DESC
            ) AS revenue_rank
        FROM ecom.products p
        JOIN ecom.order_items oi
          ON oi.product_id = p.product_id
        JOIN ecom.orders o
          ON o.order_id = oi.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY
            p.product_id,
            p.product_name
        ORDER BY
            revenue_rank,
            p.product_id
        LIMIT 20
```

튜닝 전 실행 결과

Q04 튜닝 전 결과

```
product_id | product_name | revenue      | revenue_rank
-----------+--------------+--------------+-------------
457        | Product 208  | 2,340,762.83 | 1
588        | Product 510  | 998,376.91   | 2
85         | Product 112  | 21,508.92    | 3
317        | Product 156  | 17,799.75    | 4
347        | Product 466  | 17,102.35    | 5
467        | Product 58   | 16,838.52    | 6
234        | Product 174  | 16,782.15    | 7
540        | Product 359  | 15,995.89    | 8
... 12 rows omitted
rows=20 | median execution=5.415 ms
```

튜닝 전 성능 원문 로그 · Q04_plan_before.png (1/2)

```
Q04 상품별 누적매출 RANK Top 20 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
        p.product_id,
        p.product_name,
        sum(oi.line_total) AS revenue,
        rank() OVER (
          ORDER BY sum(oi.line_total) DESC
        ) AS revenue_rank
      FROM ecom.products p
      JOIN ecom.order_items oi
        ON oi.product_id = p.product_id
      JOIN ecom.orders o
        ON o.order_id = oi.order_id
      WHERE o.order_status IN ('paid', 'shipped', 'delivered')
      GROUP BY
        p.product_id,
        p.product_name
      ORDER BY
        revenue_rank,
        p.product_id
      LIMIT 20

PERFORMANCE RESULT
Limit  (cost=1167.81..1167.86 rows=20 width=59) (actual time=15.491..15.495 rows=20 loops=1)
  Buffers: shared hit=373
  ->  Sort  (cost=1167.81..1169.31 rows=600 width=59) (actual time=15.489..15.490 rows=20 loops=1)
        Sort Key: (rank() OVER (?)), p.product_id
        Sort Method: top-N heapsort  Memory: 26kB
        Buffers: shared hit=373
        ->  WindowAgg  (cost=1141.36..1151.85 rows=600 width=59) (actual time=15.306..15.406 rows=600
loops=1)
              Buffers: shared hit=370
              ->  Sort  (cost=1141.35..1142.85 rows=600 width=51) (actual time=15.299..15.312 rows=600
loops=1)
                    Sort Key: (sum(oi.line_total)) DESC
                    Sort Method: quicksort  Memory: 53kB
                    Buffers: shared hit=370
                    ->  HashAggregate  (cost=1106.16..1113.66 rows=600 width=51) (actual
time=15.162..15.209 rows=600 loops=1)
                          Group Key: p.product_id
                          Batches: 1  Memory Usage: 297kB
                          Buffers: shared hit=367
                          ->  Hash Join  (cost=353.65..1003.33 rows=20565 width=25) (actual
time=2.766..11.424 rows=20557 loops=1)
                                Hash Cond: (oi.product_id = p.product_id)
                                Buffers: shared hit=367
                                ->  Hash Join  (cost=333.15..928.47 rows=20565 width=14) (actual
time=2.538..8.490 rows=20557 loops=1)
                                      Hash Cond: (oi.order_id = o.order_id)
                                      Buffers: shared hit=360
                                      ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716
width=22) (actual time=0.016..1.722 rows=26716 loops=1)
                                            Buffers: shared hit=258
                                      ->  Hash  (cost=237.99..237.99 rows=7613 width=8) (actual
time=2.493..2.493 rows=7613 loops=1)
                                            Buckets: 8192  Batches: 1  Memory Usage: 362kB
                                            Buffers: shared hit=102
                                            ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613
width=8) (actual time=0.008..1.746 rows=7613 loops=1)
                                                  Filter: (order_status = ANY
('{paid,shipped,delivered}'::text[]))
                                                  Rows Removed by Filter: 2277
                                                  Buffers: shared hit=102
```

튜닝 전 성능 원문 로그 · Q04_plan_before.png (2/2)

```
                                Buffers: shared hit=102
                          ->  Hash  (cost=13.00..13.00 rows=600 width=19) (actual time=0.167..0.168
rows=600 loops=1)
                                Buckets: 1024  Batches: 1  Memory Usage: 39kB
                                Buffers: shared hit=7
                                ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19)
(actual time=0.028..0.105 rows=600 loops=1)
                                      Buffers: shared hit=7
Planning:
  Buffers: shared hit=303
Planning Time: 2.003 ms
Execution Time: 15.623 ms
```

튜닝 포인트

변경 상품 매출 집계→순위→상품명 조회로 역할 분리

튜닝 후

```
WITH product_revenue AS (
            SELECT
                oi.product_id,
                sum(oi.line_total) AS revenue
            FROM ecom.order_items oi
            JOIN ecom.orders o
              ON o.order_id = oi.order_id
            WHERE o.order_status IN ('paid', 'shipped', 'delivered')
            GROUP BY oi.product_id
        ),
        ranked_products AS (
            SELECT
                product_id,
                revenue,
                rank() OVER (ORDER BY revenue DESC) AS revenue_rank
            FROM product_revenue
        )

        SELECT
            rp.product_id,
            p.product_name,
            rp.revenue,
            rp.revenue_rank
        FROM ranked_products rp
        JOIN ecom.products p
          ON p.product_id = rp.product_id
        WHERE rp.revenue_rank <= 20
        ORDER BY
            rp.revenue_rank,
            rp.product_id
        LIMIT 20
```

튜닝 후 실행 결과

Q04 튜닝 후 결과

```
product_id | product_name | revenue      | revenue_rank
-----------+--------------+--------------+-------------
457        | Product 208  | 2,340,762.83 | 1
588        | Product 510  | 998,376.91   | 2
85         | Product 112  | 21,508.92    | 3
317        | Product 156  | 17,799.75    | 4
347        | Product 466  | 17,102.35    | 5
467        | Product 58   | 16,838.52    | 6
234        | Product 174  | 16,782.15    | 7
540        | Product 359  | 15,995.89    | 8
... 12 rows omitted
rows=20 | median execution=4.014 ms
```

튜닝 후 성능 원문 로그 · Q04_plan_after.png (1/2)

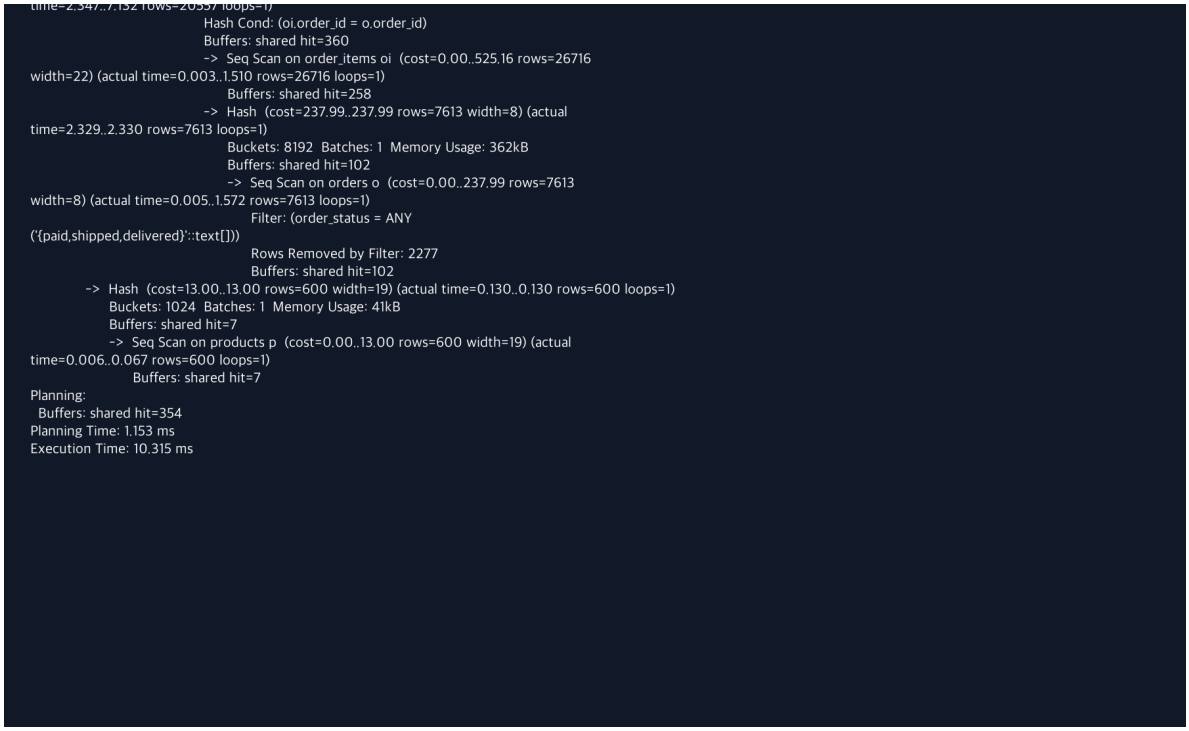
```
Q04 상품별 누적매출 RANK Top 20 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH product_revenue AS (
        SELECT
          oi.product_id,
          sum(oi.line_total) AS revenue
        FROM ecom.order_items oi
        JOIN ecom.orders o
          ON o.order_id = oi.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY oi.product_id
    ),
    ranked_products AS (
        SELECT
          product_id,
          revenue,
          rank() OVER (ORDER BY revenue DESC) AS revenue_rank
        FROM product_revenue
    )

    SELECT
        rp.product_id,
        p.product_name,
        rp.revenue,
        rp.revenue_rank
    FROM ranked_products rp
    JOIN ecom.products p
      ON p.product_id = rp.product_id
    WHERE rp.revenue_rank <= 20
    ORDER BY
        rp.revenue_rank,
        rp.product_id
    LIMIT 20

PERFORMANCE RESULT
Limit  (cost=1115.04..1115.09 rows=20 width=59) (actual time=10.185..10.193 rows=20 loops=1)
  Buffers: shared hit=373
  ->  Sort  (cost=1115.04..1116.54 rows=600 width=59) (actual time=10.184..10.187 rows=20 loops=1)
        Sort Key: (rank() OVER (?)), product_revenue.product_id
        Sort Method: quicksort  Memory: 26kB
        Buffers: shared hit=373
        ->  Hash Join  (cost=1087.00..1099.07 rows=600 width=59) (actual time=10.151..10.169 rows=20
loops=1)
              Hash Cond: (product_revenue.product_id = p.product_id)
              Buffers: shared hit=370
              ->  WindowAgg  (cost=1066.50..1076.98 rows=600 width=48) (actual time=10.014..10.026
rows=20 loops=1)
                    Run Condition: (rank() OVER (?) <= 20)
                    Buffers: shared hit=363
                    ->  Sort  (cost=1066.48..1067.98 rows=600 width=40) (actual time=10.006..10.008
rows=21 loops=1)
                          Sort Key: product_revenue.revenue DESC
                          Sort Method: quicksort  Memory: 44kB
                          Buffers: shared hit=363
                          ->  Subquery Scan on product_revenue  (cost=1031.30..1038.80 rows=600 width=40)
(actual time=9.716..9.851 rows=600 loops=1)
                                Buffers: shared hit=360
                                ->  HashAggregate  (cost=1031.30..1038.80 rows=600 width=40) (actual
time=9.715..9.805 rows=600 loops=1)
                                      Group Key: oi.product_id
                                      Batches: 1  Memory Usage: 297kB
                                      Buffers: shared hit=360
                                      ->  Hash Join  (cost=333.15..928.47 rows=20565 width=14) (actual
time=2.247..7.122 rows=20557 loops=1)
```

튜닝 후 성능 원문 로그 · Q04_plan_after.png (2/2)



```
time=2.347..7.132 rows=20537 loops=1)
                                  Hash Cond: (oi.order_id = o.order_id)
                                  Buffers: shared hit=360
                                  ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716
width=22) (actual time=0.003..1.510 rows=26716 loops=1)
                                        Buffers: shared hit=258
                                  ->  Hash  (cost=237.99..237.99 rows=7613 width=8) (actual
time=2.329..2.330 rows=7613 loops=1)
                                        Buckets: 8192  Batches: 1  Memory Usage: 362kB
                                        Buffers: shared hit=102
                                        ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613
width=8) (actual time=0.005..1.572 rows=7613 loops=1)
                                              Filter: (order_status = ANY
('{paid,shipped,delivered}'::text[]))
                                              Rows Removed by Filter: 2277
                                              Buffers: shared hit=102
            ->  Hash  (cost=13.00..13.00 rows=600 width=19) (actual time=0.130..0.130 rows=600 loops=1)
                  Buckets: 1024  Batches: 1  Memory Usage: 41kB
                  Buffers: shared hit=7
                  ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19) (actual
time=0.006..0.067 rows=600 loops=1)
                        Buffers: shared hit=7
Planning:
  Buffers: shared hit=354
Planning Time: 1.153 ms
Execution Time: 10.315 ms
```

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	5.415 ms	4.014 ms
Planning Time	0.092 ms	0.065 ms
Actual Rows	20	20
Shared Hit	367	367
Shared Read	0	0
Top Plan Node	Limit	Limit
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

정확히 20개를 반환하고 동점은 product_id로 결정했다.

MD 전달안

누적매출 1위는 Product 208(ID 457), 매출 2,340,762.83다. 결과는 정확히 20개로 고정했다.

Q5. 고객 RFM

요청 부서	CRM
요청 목적	고객별 최근성·빈도·금액 파악
판정 기준	유효 주문·주문 단위 빈도

튜닝 전

주문상품 행에서 COUNT DISTINCT로 빈도 복원

```
SELECT
            c.customer_id,
            c.full_name,
            current_date - max(o.order_ts)::date AS recency_days,
            count(DISTINCT o.order_id) AS frequency,
            sum(oi.line_total) AS monetary
        FROM ecom.customers c
        JOIN ecom.orders o
          ON o.customer_id = c.customer_id
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY
            c.customer_id,
            c.full_name
        ORDER BY
            monetary DESC,
            c.customer_id
```

튜닝 전 실행 결과

Q05 튜닝 전 결과

```
customer_id | full_name   | recency_days | frequency | monetary
------------+-------------+--------------+-----------+----------
19          | Customer 19 | 1            | 26        | 37,803.69
7           | Customer 7  | 6            | 23        | 36,318.34
3           | Customer 3  | 3            | 26        | 33,809.53
25          | Customer 25 | 4            | 26        | 33,689.05
1           | Customer 1  | 3            | 26        | 33,226.61
10          | Customer 10 | 1            | 26        | 33,080.74
21          | Customer 21 | 1            | 26        | 32,305.26
15          | Customer 15 | 8            | 26        | 31,575.76
... 2,242 rows omitted
rows=2,250 | median execution=8.810 ms
```

튜닝 전 성능 원문 로그 · Q05_plan_before.png (1/2)

```
Q05 고객 RFM [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
        c.customer_id,
        c.full_name,
        current_date - max(o.order_ts)::date AS recency_days,
        count(DISTINCT o.order_id) AS frequency,
        sum(oi.line_total) AS monetary
    FROM ecom.customers c
    JOIN ecom.orders o
      ON o.customer_id = c.customer_id
    JOIN ecom.order_items oi
      ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY
        c.customer_id,
        c.full_name
    ORDER BY
        monetary DESC,
        c.customer_id

PERFORMANCE RESULT
Sort  (cost=3053.62..3061.12 rows=3000 width=65) (actual time=16.944..17.022 rows=2250 loops=1)
  Sort Key: (sum(oi.line_total)) DESC, c.customer_id
  Sort Method: quicksort  Memory: 237kB
  Buffers: shared hit=406
  ->  GroupAggregate  (cost=2563.30..2880.36 rows=3000 width=65) (actual time=13.422..16.460 rows=2250
loops=1)
        Group Key: c.customer_id
        Buffers: shared hit=400
        ->  Sort  (cost=2563.30..2614.71 rows=20565 width=43) (actual time=13.391..13.966 rows=20557
loops=1)
              Sort Key: c.customer_id, o.order_id
              Sort Method: quicksort  Memory: 2053kB
              Buffers: shared hit=400
              ->  Hash Join  (cost=440.65..1090.03 rows=20565 width=43) (actual time=1.915..9.393
rows=20557 loops=1)
                    Hash Cond: (o.customer_id = c.customer_id)
                    Buffers: shared hit=400
                    ->  Hash Join  (cost=333.15..928.47 rows=20565 width=30) (actual time=1.399..6.478
rows=20557 loops=1)
                          Hash Cond: (oi.order_id = o.order_id)
                          Buffers: shared hit=360
                          ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.002..1.540 rows=26716 loops=1)
                                Buffers: shared hit=258
                          ->  Hash  (cost=237.99..237.99 rows=7613 width=24) (actual time=1.388..1.388
rows=7613 loops=1)
                                Buckets: 8192  Batches: 1  Memory Usage: 481kB
                                Buffers: shared hit=102
                                ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=24) (actual
time=0.002..0.967 rows=7613 loops=1)
                                      Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                      Rows Removed by Filter: 2277
                                      Buffers: shared hit=102
                    ->  Hash  (cost=70.00..70.00 rows=3000 width=21) (actual time=0.509..0.509 rows=3000
loops=1)
                          Buckets: 4096  Batches: 1  Memory Usage: 190kB
                          Buffers: shared hit=40
                          ->  Seq Scan on customers c  (cost=0.00..70.00 rows=3000 width=21) (actual
time=0.007..0.279 rows=3000 loops=1)
                                Buffers: shared hit=40
Planning:
  Buffers: shared hit=339
```

튜닝 전 성능 원문 로그 · Q05_plan_before.png (2/2)

```
Planning Time: 1.202 ms
Execution Time: 17.166 ms
```

튜닝 포인트

변경 주문별 금액을 선집계한 뒤 고객 RFM 계산

튜닝 후

```
WITH order_totals AS (
            SELECT
                o.customer_id,
                o.order_id,
                o.order_ts,
                sum(oi.line_total) AS amount
            FROM ecom.orders o
            JOIN ecom.order_items oi
              ON oi.order_id = o.order_id
            WHERE o.order_status IN ('paid', 'shipped', 'delivered')
            GROUP BY
                o.customer_id,
                o.order_id,
                o.order_ts
        )

        SELECT
            c.customer_id,
            c.full_name,
            current_date - max(ot.order_ts)::date AS recency_days,
            count(*) AS frequency,
            sum(ot.amount) AS monetary
        FROM order_totals ot
        JOIN ecom.customers c
          ON c.customer_id = ot.customer_id
        GROUP BY
            c.customer_id,
            c.full_name
        ORDER BY
            monetary DESC,
            c.customer_id
```

튜닝 후 실행 결과

Q05 튜닝 후 결과

```
customer_id | full_name   | recency_days | frequency | monetary
------------+-------------+--------------+-----------+----------
19          | Customer 19 | 1            | 26        | 37,803.69
7           | Customer 7  | 6            | 23        | 36,318.34
3           | Customer 3  | 3            | 26        | 33,809.53
25          | Customer 25 | 4            | 26        | 33,689.05
1           | Customer 1  | 3            | 26        | 33,226.61
10          | Customer 10 | 1            | 26        | 33,080.74
21          | Customer 21 | 1            | 26        | 32,305.26
15          | Customer 15 | 8            | 26        | 31,575.76
... 2,242 rows omitted
rows=2,250 | median execution=6.844 ms
```

튜닝 후 성능 원문 로그 · Q05_plan_after.png (1/2)

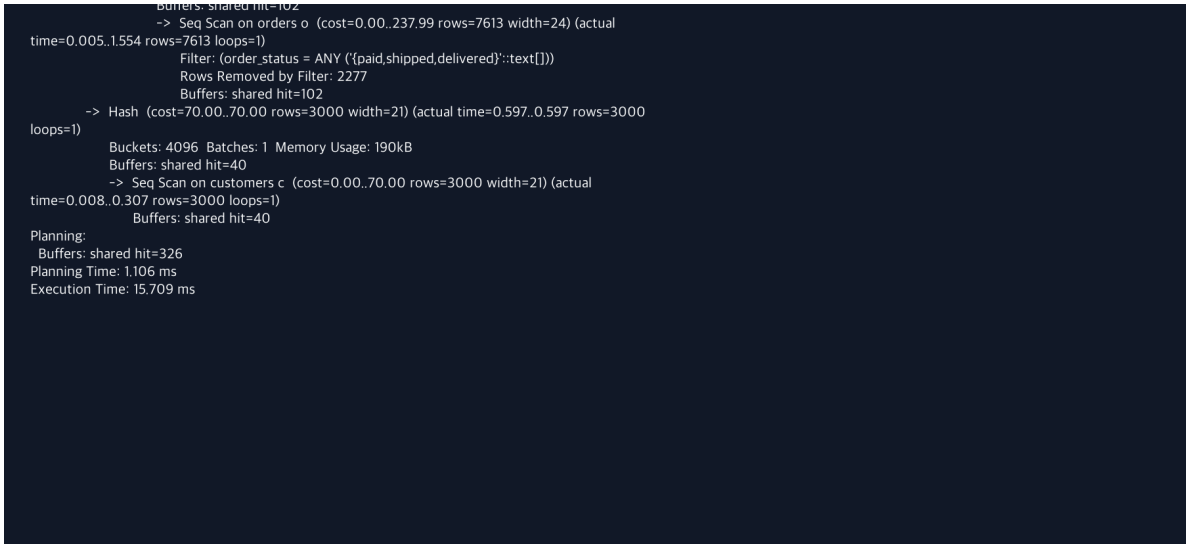
```
Q05 고객 RFM [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH order_totals AS (
        SELECT
          o.customer_id,
          o.order_id,
          o.order_ts,
          sum(oi.line_total) AS amount
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY
          o.customer_id,
          o.order_id,
          o.order_ts
    )

    SELECT
      c.customer_id,
      c.full_name,
      current_date - max(ot.order_ts)::date AS recency_days,
      count(*) AS frequency,
      sum(ot.amount) AS monetary
    FROM order_totals ot
    JOIN ecom.customers c
      ON c.customer_id = ot.customer_id
    GROUP BY
      c.customer_id,
      c.full_name
    ORDER BY
      monetary DESC,
      c.customer_id

PERFORMANCE RESULT
Sort  (cost=1639.50..1647.00 rows=3000 width=65) (actual time=15.477..15.527 rows=2250 loops=1)
  Sort Key: (sum((sum(oi.line_total)))) DESC, c.customer_id
  Sort Method: quicksort  Memory: 237kB
  Buffers: shared hit=406
  ->  HashAggregate  (cost=1406.24..1466.24 rows=3000 width=65) (actual time=14.218..14.901 rows=2250
loops=1)
      Group Key: c.customer_id
      Batches: 1  Memory Usage: 1137kB
      Buffers: shared hit=400
      ->  Hash Join  (cost=1138.80..1330.11 rows=7613 width=61) (actual time=10.028..12.756 rows=7613
loops=1)
          Hash Cond: (o.customer_id = c.customer_id)
          Buffers: shared hit=400
          ->  HashAggregate  (cost=1031.30..1126.46 rows=7613 width=56) (actual time=9.419..10.999
rows=7613 loops=1)
              Group Key: o.order_id
              Batches: 1  Memory Usage: 3473kB
              Buffers: shared hit=360
              ->  Hash Join  (cost=333.15..928.47 rows=20565 width=30) (actual time=2.384..6.628
rows=20557 loops=1)
                  Hash Cond: (oi.order_id = o.order_id)
                  Buffers: shared hit=360
                  ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.003..1.193 rows=26716 loops=1)
                      Buffers: shared hit=258
                  ->  Hash  (cost=237.99..237.99 rows=7613 width=24) (actual time=2.360..2.361
rows=7613 loops=1)
                      Buckets: 8192  Batches: 1  Memory Usage: 481kB
                      Buffers: shared hit=102
```

튜닝 후 성능 원문 로그 · Q05_plan_after.png (2/2)



```
                        Buffers: shared hit=102
                        ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=24) (actual
time=0.005..1.554 rows=7613 loops=1)
                              Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                              Rows Removed by Filter: 2277
                              Buffers: shared hit=102
          ->  Hash  (cost=70.00..70.00 rows=3000 width=21) (actual time=0.597..0.597 rows=3000
loops=1)
                Buckets: 4096  Batches: 1  Memory Usage: 190kB
                Buffers: shared hit=40
                ->  Seq Scan on customers c  (cost=0.00..70.00 rows=3000 width=21) (actual
time=0.008..0.307 rows=3000 loops=1)
                      Buffers: shared hit=40
Planning:
  Buffers: shared hit=326
Planning Time: 1.106 ms
Execution Time: 15.709 ms
```

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	8.810 ms	6.844 ms
Planning Time	0.079 ms	0.068 ms
Actual Rows	2,250	2,250
Shared Hit	400	400
Shared Read	0	0
Top Plan Node	Sort	Sort
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

빈도와 금액의 집계 단위를 주문으로 통일했다.

CRM 전달안

최고 Monetary 고객은 Customer 19(ID 19)이며 구매 26회, 누적금액 37,803.69, 최근성 1일이다.

Q6. 첫 구매 후 30일 내 재구매율

요청 부서	CRM
요청 목적	첫 구매 후 30일 내 재구매율 확인
판정 기준	첫 주문 이후 추가 주문

튜닝 전

고객별 EXISTS 상관 서브쿼리를 반복

```
WITH purchases AS (
            SELECT
                customer_id,
                order_ts
            FROM ecom.orders
            WHERE order_status IN ('paid', 'shipped', 'delivered')
        ),
        first_buy AS (
            SELECT
                customer_id,
                min(order_ts) AS first_order_ts
            FROM purchases
            GROUP BY customer_id
        )

        SELECT
            count(*) AS first_buyers,
            count(*) FILTER (
                WHERE EXISTS (
                    SELECT 1
                    FROM purchases p
                    WHERE p.customer_id = f.customer_id
                      AND p.order_ts > f.first_order_ts
                      AND p.order_ts <= f.first_order_ts + interval '30 days'
                )
            ) AS repurchasers,
            ecom.safe_div(
                count(*) FILTER (
                    WHERE EXISTS (
                        SELECT 1
                        FROM purchases p
                        WHERE p.customer_id = f.customer_id
                          AND p.order_ts > f.first_order_ts
                          AND p.order_ts <= f.first_order_ts + interval '30 days'
                    )
                ),
                count(*)
            ) AS repurchase_rate
        FROM first_buy f
```

튜닝 전 실행 결과

Q06 튜닝 전 결과

```
first_buyers | repurchasers | repurchase_rate
-------------+--------------+----------------
2250         | 1378         | 0.61
rows=1 | median execution=514.634 ms
```

튜닝 전 성능 원문 로그 · Q06_plan_before.png (1/2)

```
Q06 첫 구매 후 30일 내 재구매율 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH purchases AS (
        SELECT
            customer_id,
            order_ts
        FROM ecom.orders
        WHERE order_status IN ('paid', 'shipped', 'delivered')
    ),
    first_buy AS (
        SELECT
            customer_id,
            min(order_ts) AS first_order_ts
        FROM purchases
        GROUP BY customer_id
    )

    SELECT
        count(*) AS first_buyers,
        count(*) FILTER (
            WHERE EXISTS (
                SELECT 1
                FROM purchases p
                WHERE p.customer_id = f.customer_id
                  AND p.order_ts > f.first_order_ts
                  AND p.order_ts <= f.first_order_ts + interval '30 days'
            )
        ) AS repurchasers,
        ecom.safe_div(
            count(*) FILTER (
                WHERE EXISTS (
                    SELECT 1
                    FROM purchases p
                    WHERE p.customer_id = f.customer_id
                      AND p.order_ts > f.first_order_ts
                      AND p.order_ts <= f.first_order_ts + interval '30 days'
                )
            ),
            count(*)
        ) AS repurchase_rate
    FROM first_buy f

PERFORMANCE RESULT
Aggregate  (cost=1055173.74..1055173.76 rows=1 width=48) (actual time=710.017..710.021 rows=1 loops=1)
  Buffers: shared hit=102
  CTE purchases
    ->  Seq Scan on orders  (cost=0.00..237.99 rows=7613 width=16) (actual time=0.004..0.668 rows=7613
loops=1)
          Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
          Rows Removed by Filter: 2277
          Buffers: shared hit=102
  ->  HashAggregate  (cost=190.32..213.41 rows=2309 width=16) (actual time=1.505..1.847 rows=2250
loops=1)
        Group Key: purchases.customer_id
        Batches: 1  Memory Usage: 369kB
        Buffers: shared hit=102
        ->  CTE Scan on purchases  (cost=0.00..152.26 rows=7613 width=16) (actual time=0.005..1.098
rows=7613 loops=1)
              Buffers: shared hit=102
  SubPlan 2
    ->  CTE Scan on purchases p  (cost=0.00..228.39 rows=1 width=0) (actual time=0.163..0.163 rows=1
loops=2250)
          Filter: ((order_ts > (min(purchases.order_ts))) AND (customer_id = purchases.customer_id) AND
(order_ts <= ((min(purchases.order_ts)) + '30 days'::interval)))
```

튜닝 전 성능 원문 로그 · Q06_plan_before.png (2/2)

```
(order_ts <= ((min(purchases.order_ts)) + '30 days'::interval)))
        Rows Removed by Filter: 4963
  SubPlan 3
    ->  CTE Scan on purchases p_1  (cost=0.00..228.39 rows=1 width=0) (actual time=0.151..0.151 rows=1
loops=2250)
        Filter: ((order_ts > (min(purchases.order_ts))) AND (customer_id = purchases.customer_id) AND
(order_ts <= ((min(purchases.order_ts)) + '30 days'::interval)))
        Rows Removed by Filter: 4963
Planning:
  Buffers: shared hit=408
Planning Time: 0.424 ms
Execution Time: 710.099 ms
```

튜닝 포인트

변경 윈도 함수로 첫 구매일 계산 후 BOOL_OR로 고객당 1회 집계

튜닝 후

```
WITH purchases AS (
            SELECT
                customer_id,
                order_ts,
                min(order_ts) OVER (
                    PARTITION BY customer_id
                ) AS first_order_ts
            FROM ecom.orders
            WHERE order_status IN ('paid', 'shipped', 'delivered')
        ),
        customer_flags AS (
            SELECT
                customer_id,
                bool_or(
                    order_ts > first_order_ts
                    AND order_ts <= first_order_ts + interval '30 days'
                ) AS repurchased
            FROM purchases
            GROUP BY customer_id
        )

        SELECT
            count(*) AS first_buyers,
            count(*) FILTER (WHERE repurchased) AS repurchasers,
            ecom.safe_div(
                count(*) FILTER (WHERE repurchased),
                count(*)
            ) AS repurchase_rate
        FROM customer_flags
```

튜닝 후 실행 결과

Q06 튜닝 후 결과

```
first_buyers | repurchasers | repurchase_rate
-------------+--------------+----------------
2250         | 1378         | 0.61
rows=1 | median execution=2.702 ms
```

튜닝 후 성능 원문 로그 · Q06_plan_after.png (1/2)

```
Q06 첫 구매 후 30일 내 재구매율 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH purchases AS (
        SELECT
            customer_id,
            order_ts,
            min(order_ts) OVER (
                PARTITION BY customer_id
            ) AS first_order_ts
        FROM ecom.orders
        WHERE order_status IN ('paid', 'shipped', 'delivered')
    ),
    customer_flags AS (
        SELECT
            customer_id,
            bool_or(
                order_ts > first_order_ts
                AND order_ts <= first_order_ts + interval '30 days'
            ) AS repurchased
        FROM purchases
        GROUP BY customer_id
    )

    SELECT
        count(*) AS first_buyers,
        count(*) FILTER (WHERE repurchased) AS repurchasers,
        ecom.safe_div(
            count(*) FILTER (WHERE repurchased),
            count(*)
        ) AS repurchase_rate
    FROM customer_flags

PERFORMANCE RESULT
Aggregate  (cost=1014.92..1014.95 rows=1 width=48) (actual time=2.957..2.959 rows=1 loops=1)
  Buffers: shared hit=105
  ->  GroupAggregate  (cost=728.84..980.29 rows=2309 width=9) (actual time=1.097..2.902 rows=2250
loops=1)
        Group Key: orders.customer_id
        Buffers: shared hit=105
        ->  WindowAgg  (cost=728.84..862.03 rows=7613 width=24) (actual time=1.086..2.016 rows=7613
loops=1)
              Buffers: shared hit=105
              ->  Sort  (cost=728.81..747.84 rows=7613 width=16) (actual time=1.080..1.210 rows=7613
loops=1)
                    Sort Key: orders.customer_id
                    Sort Method: quicksort  Memory: 430kB
                    Buffers: shared hit=105
                    ->  Seq Scan on orders  (cost=0.00..237.99 rows=7613 width=16) (actual
time=0.002..0.650 rows=7613 loops=1)
                          Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                          Rows Removed by Filter: 2277
                          Buffers: shared hit=102
Planning:
  Buffers: shared hit=418
Planning Time: 0.288 ms
Execution Time: 2.983 ms
```

튜닝 후 성능 원문 로그 · Q06_plan_after.png (2/2)

```
```

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	514.634 ms	2.702 ms
Planning Time	0.085 ms	0.034 ms
Actual Rows	1	1
Shared Hit	102	102
Shared Read	0	0
Top Plan Node	Aggregate	Aggregate
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

반복 탐색을 단일 스캔·집계 흐름으로 바꾼 대표 개선 사례다.

CRM 전달안

첫 구매 고객 2,250명 중 1,378명이 30일 내 재구매해 재구매율은 61.2%다.

Q7. 재고 부족 위험 상품

요청 부서	구매·SCM
요청 목적	재주문이 필요한 상품 탐지
판정 기준	현재고 < 재주문 임계치

튜닝 전

상품마다 재고 스칼라 서브쿼리를 네 번 실행

```
SELECT
            p.product_id,
            p.product_name,
            (
                SELECT i.qty_on_hand
                FROM ecom.inventory i
                WHERE i.product_id = p.product_id
            ) AS qty_on_hand,
            (
                SELECT i.reorder_point
                FROM ecom.inventory i
                WHERE i.product_id = p.product_id
            ) AS reorder_point,
            (
                SELECT i.reorder_point - i.qty_on_hand
                FROM ecom.inventory i
                WHERE i.product_id = p.product_id
            ) AS shortage
        FROM ecom.products p
        WHERE (
            SELECT i.qty_on_hand < i.reorder_point
            FROM ecom.inventory i
            WHERE i.product_id = p.product_id
        )
        ORDER BY
            shortage DESC,
            p.product_id
```

튜닝 전 실행 결과

Q07 튜닝 전 결과

```
product_id | product_name | qty_on_hand | reorder_point | shortage
-----------+--------------+-------------+---------------+---------
408        | Product 37   | 0           | 30            | 30
427        | Product 138  | 0           | 30            | 30
567        | Product 20   | 0           | 30            | 30
165        | Product 363  | 1           | 30            | 29
239        | Product 84   | 1           | 30            | 29
378        | Product 357  | 1           | 30            | 29
457        | Product 208  | 1           | 30            | 29
547        | Product 470  | 1           | 30            | 29
... 55 rows omitted
rows=63 | median execution=0.321 ms
```

튜닝 전 성능 원문 로그 · Q07_plan_before.png (1/2)

```
Q07 재고 부족 위험 상품 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
        p.product_id,
        p.product_name,
        (
          SELECT i.qty_on_hand
          FROM ecom.inventory i
          WHERE i.product_id = p.product_id
        ) AS qty_on_hand,
        (
          SELECT i.reorder_point
          FROM ecom.inventory i
          WHERE i.product_id = p.product_id
        ) AS reorder_point,
        (
          SELECT i.reorder_point - i.qty_on_hand
          FROM ecom.inventory i
          WHERE i.product_id = p.product_id
        ) AS shortage
      FROM ecom.products p
      WHERE (
        SELECT i.qty_on_hand < i.reorder_point
        FROM ecom.inventory i
        WHERE i.product_id = p.product_id
      )
      ORDER BY
        shortage DESC,
        p.product_id

PERFORMANCE RESULT
Sort  (cost=12466.34..12467.09 rows=300 width=31) (actual time=0.908..0.913 rows=63 loops=1)
  Sort Key: ((SubPlan 3)) DESC, p.product_id
  Sort Method: quicksort  Memory: 27kB
  Buffers: shared hit=2380
  ->  Seq Scan on products p  (cost=0.00..12454.00 rows=300 width=31) (actual time=0.038..0.872 rows=63
loops=1)
        Filter: (SubPlan 4)
        Rows Removed by Filter: 537
        Buffers: shared hit=2374
        SubPlan 1
          ->  Index Scan using inventory_pkey on inventory i  (cost=0.28..8.29 rows=1 width=4) (actual
time=0.001..0.001 rows=1 loops=63)
                Index Cond: (product_id = p.product_id)
                Buffers: shared hit=189
        SubPlan 2
          ->  Index Scan using inventory_pkey on inventory i_1  (cost=0.28..8.29 rows=1 width=4) (actual
time=0.001..0.001 rows=1 loops=63)
                Index Cond: (product_id = p.product_id)
                Buffers: shared hit=189
        SubPlan 3
          ->  Index Scan using inventory_pkey on inventory i_2  (cost=0.28..8.29 rows=1 width=4) (actual
time=0.001..0.001 rows=1 loops=63)
                Index Cond: (product_id = p.product_id)
                Buffers: shared hit=189
        SubPlan 4
          ->  Index Scan using inventory_pkey on inventory i_3  (cost=0.28..8.29 rows=1 width=1) (actual
time=0.001..0.001 rows=1 loops=600)
                Index Cond: (product_id = p.product_id)
                Buffers: shared hit=1800
Planning:
  Buffers: shared hit=171
Planning Time: 0.637 ms
Execution Time: 0.964 ms
```

튜닝 전 성능 원문 로그 · Q07_plan_before.png (2/2)

튜닝 포인트

변경 inventory와 products를 한 번 JOIN

튜닝 후

```
SELECT
            p.product_id,
            p.product_name,
            i.qty_on_hand,
            i.reorder_point,
            i.reorder_point - i.qty_on_hand AS shortage
        FROM ecom.inventory i
        JOIN ecom.products p
          ON p.product_id = i.product_id
        WHERE i.qty_on_hand < i.reorder_point
        ORDER BY
            shortage DESC,
            p.product_id
```

튜닝 후 실행 결과

Q07 튜닝 후 결과

```
product_id | product_name | qty_on_hand | reorder_point | shortage
-----------+--------------+-------------+---------------+---------
408        | Product 37   | 0           | 30            | 30
427        | Product 138  | 0           | 30            | 30
567        | Product 20   | 0           | 30            | 30
165        | Product 363  | 1           | 30            | 29
239        | Product 84   | 1           | 30            | 29
378        | Product 357  | 1           | 30            | 29
457        | Product 208  | 1           | 30            | 29
547        | Product 470  | 1           | 30            | 29
... 55 rows omitted
rows=63 | median execution=0.070 ms
```

튜닝 후 성능 원문 로그 · Q07_plan_after.png

```
Q07 재고 부족 위험 상품 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
        p.product_id,
        p.product_name,
        i.qty_on_hand,
        i.reorder_point,
        i.reorder_point - i.qty_on_hand AS shortage
    FROM ecom.inventory i
    JOIN ecom.products p
      ON p.product_id = i.product_id
    WHERE i.qty_on_hand < i.reorder_point
    ORDER BY
        shortage DESC,
        p.product_id

PERFORMANCE RESULT
Sort  (cost=37.73..38.23 rows=200 width=31) (actual time=0.297..0.305 rows=63 loops=1)
  Sort Key: ((i.reorder_point - i.qty_on_hand)) DESC, p.product_id
  Sort Method: quicksort  Memory: 27kB
  Buffers: shared hit=18
  ->  Hash Join  (cost=15.00..30.09 rows=200 width=31) (actual time=0.125..0.254 rows=63 loops=1)
        Hash Cond: (p.product_id = i.product_id)
        Buffers: shared hit=12
        ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19) (actual time=0.007..0.064
rows=600 loops=1)
              Buffers: shared hit=7
        ->  Hash  (cost=12.50..12.50 rows=200 width=16) (actual time=0.108..0.108 rows=63 loops=1)
              Buckets: 1024  Batches: 1  Memory Usage: 11kB
              Buffers: shared hit=5
              ->  Seq Scan on inventory i  (cost=0.00..12.50 rows=200 width=16) (actual time=0.014..0.086
rows=63 loops=1)
                    Filter: (qty_on_hand < reorder_point)
                    Rows Removed by Filter: 537
                    Buffers: shared hit=5
Planning:
  Buffers: shared hit=199
Planning Time: 1.128 ms
Execution Time: 0.368 ms
```

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	0.321 ms	0.070 ms
Planning Time	0.022 ms	0.032 ms
Actual Rows	63	63
Shared Hit	2,374	12
Shared Read	0	0
Top Plan Node	Sort	Sort
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

같은 테이블 반복 조회를 제거했다.

구매·SCM 전달안

재고 임계치 미달 상품은 63개다. 가장 긴급한 Product 37(ID 408)은 현재고 0, 임계치 30, 부족분 30다.

Q8. 리뷰 기반 효자상품

요청 부서	MD·마케팅
요청 목적	리뷰 기반 효자상품 선별
판정 기준	평점 4.5 이상·리뷰 50개 이상

튜닝 전

상품명까지 조인한 뒤 모든 리뷰를 집계

```
SELECT
            p.product_id,
            p.product_name,
            avg(r.rating) AS avg_rating,
            count(*) AS review_count
        FROM ecom.products p
        JOIN ecom.reviews r
          ON r.product_id = p.product_id
        GROUP BY
            p.product_id,
            p.product_name
        HAVING avg(r.rating) >= 4.5
           AND count(*) >= 50
        ORDER BY
            avg_rating DESC,
            review_count DESC,
            p.product_id
```

튜닝 전 실행 결과

Q08 튜닝 전 결과

```
product_id | product_name | avg_rating | review_count
-----------+--------------+------------+-------------
418        | Product 297  | 4.86       | 160
544        | Product 260  | 4.81       | 161
166        | Product 83   | 4.80       | 160
267        | Product 235  | 4.79       | 160
490        | Product 259  | 4.79       | 160
552        | Product 60   | 4.78       | 161
341        | Product 406  | 4.77       | 160
468        | Product 268  | 4.76       | 160
... 4 rows omitted
rows=12 | median execution=0.441 ms
```

튜닝 전 성능 원문 로그 · Q08_plan_before.png

```
Q08 리뷰 기반 효자상품 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
        p.product_id,
        p.product_name,
        avg(r.rating) AS avg_rating,
        count(*) AS review_count
    FROM ecom.products p
    JOIN ecom.reviews r
      ON r.product_id = p.product_id
    GROUP BY
        p.product_id,
        p.product_name
    HAVING avg(r.rating) >= 4.5
       AND count(*) >= 50
    ORDER BY
        avg_rating DESC,
        review_count DESC,
        p.product_id

PERFORMANCE RESULT
Sort  (cost=95.24..95.41 rows=67 width=59) (actual time=1.308..1.313 rows=12 loops=1)
  Sort Key: (avg(r.rating)) DESC, (count(*)) DESC, p.product_id
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=34
  ->  HashAggregate  (cost=82.71..93.21 rows=67 width=59) (actual time=1.229..1.274 rows=12 loops=1)
        Group Key: p.product_id
        Filter: ((avg(r.rating) >= 4.5) AND (count(*) >= 50))
        Batches: 1  Memory Usage: 73kB
        Rows Removed by Filter: 123
        Buffers: shared hit=28
        ->  Hash Join  (cost=20.50..67.36 rows=2046 width=23) (actual time=0.198..0.825 rows=2046
loops=1)
              Hash Cond: (r.product_id = p.product_id)
              Buffers: shared hit=28
              ->  Seq Scan on reviews r  (cost=0.00..41.46 rows=2046 width=12) (actual time=0.004..0.171
rows=2046 loops=1)
                    Buffers: shared hit=21
              ->  Hash  (cost=13.00..13.00 rows=600 width=19) (actual time=0.187..0.187 rows=600 loops=1)
                    Buckets: 1024  Batches: 1  Memory Usage: 39kB
                    Buffers: shared hit=7
                    ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19) (actual
time=0.004..0.090 rows=600 loops=1)
                          Buffers: shared hit=7
Planning:
  Buffers: shared hit=228
Planning Time: 1.117 ms
Execution Time: 1.387 ms
```

튜닝 포인트

변경 리뷰를 상품별로 먼저 집계·필터 후 상품명 조회

튜닝 후

```
WITH review_summary AS (
            SELECT
                product_id,
                avg(rating) AS avg_rating,
                count(*) AS review_count
            FROM ecom.reviews
            GROUP BY product_id
            HAVING avg(rating) >= 4.5
               AND count(*) >= 50
        )

        SELECT
            rs.product_id,
            p.product_name,
            rs.avg_rating,
            rs.review_count
        FROM review_summary rs
        JOIN ecom.products p
          ON p.product_id = rs.product_id
        ORDER BY
            rs.avg_rating DESC,
            rs.review_count DESC,
            rs.product_id
```

튜닝 후 실행 결과

Q08 튜닝 후 결과

```
product_id | product_name | avg_rating | review_count
-----------+--------------+------------+-------------
418        | Product 297  | 4.86       | 160
544        | Product 260  | 4.81       | 161
166        | Product 83   | 4.80       | 160
267        | Product 235  | 4.79       | 160
490        | Product 259  | 4.79       | 160
552        | Product 60   | 4.78       | 161
341        | Product 406  | 4.77       | 160
468        | Product 268  | 4.76       | 160
... 4 rows omitted
rows=12 | median execution=0.274 ms
```

튜닝 후 성능 원문 로그 · Q08_plan_after.png (1/2)

```
Q08 리뷰 기반 효자상품 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH review_summary AS (
        SELECT
            product_id,
            avg(rating) AS avg_rating,
            count(*) AS review_count
        FROM ecom.reviews
        GROUP BY product_id
        HAVING avg(rating) >= 4.5
          AND count(*) >= 50
    )

    SELECT
        rs.product_id,
        p.product_name,
        rs.avg_rating,
        rs.review_count
    FROM review_summary rs
    JOIN ecom.products p
      ON p.product_id = rs.product_id
    ORDER BY
        rs.avg_rating DESC,
        rs.review_count DESC,
        rs.product_id

PERFORMANCE RESULT
Sort  (cost=74.23..74.27 rows=15 width=59) (actual time=1.023..1.028 rows=12 loops=1)
  Sort Key: (avg(reviews.rating)) DESC, (count(*)) DESC, reviews.product_id
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=34
  ->  Hash Join  (cost=59.35..73.94 rows=15 width=59) (actual time=0.854..0.983 rows=12 loops=1)
        Hash Cond: (p.product_id = reviews.product_id)
        Buffers: shared hit=28
        ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19) (actual time=0.008..0.076
rows=600 loops=1)
              Buffers: shared hit=7
        ->  Hash  (cost=59.17..59.17 rows=15 width=48) (actual time=0.834..0.834 rows=12 loops=1)
              Buckets: 1024  Batches: 1  Memory Usage: 9kB
              Buffers: shared hit=21
              ->  HashAggregate  (cost=56.80..59.17 rows=15 width=48) (actual time=0.775..0.824 rows=12
loops=1)
                    Group Key: reviews.product_id
                    Filter: ((avg(reviews.rating) >= 4.5) AND (count(*) >= 50))
                    Batches: 1  Memory Usage: 64kB
                    Rows Removed by Filter: 123
                    Buffers: shared hit=21
                    ->  Seq Scan on reviews  (cost=0.00..41.46 rows=2046 width=12) (actual
time=0.005..0.238 rows=2046 loops=1)
                          Buffers: shared hit=21
Planning:
  Buffers: shared hit=209
Planning Time: 0.974 ms
Execution Time: 1.180 ms
```

튜닝 후 성능 원문 로그 · Q08_plan_after.png (2/2)

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	0.441 ms	0.274 ms
Planning Time	0.061 ms	0.029 ms
Actual Rows	12	12
Shared Hit	28	28
Shared Read	0	0
Top Plan Node	Sort	Sort
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

효자 후보만 마스터 데이터와 결합한다.

MD·마케팅 전달안

효자상품 조건을 충족한 상품은 12개다. 상위 Product 297(ID 418)은 평점 4.86, 리뷰 160개다.

Q9. 쿠폰 사용 여부별 평균 주문 금액

요청 부서	마케팅·재무
요청 목적	쿠폰 사용 여부별 AOV 비교
판정 기준	주문별 금액을 먼저 집계

튜닝 전

주문상품 행 합계÷COUNT DISTINCT로 AOV 계산

```
SELECT
            o.coupon_code IS NOT NULL AS used_coupon,
            count(DISTINCT o.order_id) AS order_count,
            sum(oi.line_total) / count(DISTINCT o.order_id) AS avg_order_amount
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY o.coupon_code IS NOT NULL
        ORDER BY used_coupon
```

튜닝 전 실행 결과

Q09 튜닝 전 결과

```
used_coupon | order_count | avg_order_amount
------------+-------------+-----------------
False       | 5796        | 628.88
True        | 1817        | 1,839.69
rows=2 | median execution=8.061 ms
```

튜닝 전 성능 원문 로그 · Q09_plan_before.png

```
Q09 쿠폰 사용 여부별 평균 주문 금액 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
SELECT
        o.coupon_code IS NOT NULL AS used_coupon,
        count(DISTINCT o.order_id) AS order_count,
        sum(oi.line_total) / count(DISTINCT o.order_id) AS avg_order_amount
    FROM ecom.orders o
    JOIN ecom.order_items oi
      ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY o.coupon_code IS NOT NULL
    ORDER BY used_coupon

PERFORMANCE RESULT
GroupAggregate  (cost=2401.74..2607.42 rows=2 width=41) (actual time=7.779..8.156 rows=2 loops=1)
  Group Key: ((o.coupon_code IS NOT NULL))
  Buffers: shared hit=368
  ->  Sort  (cost=2401.74..2453.15 rows=20565 width=15) (actual time=6.905..7.259 rows=20557 loops=1)
        Sort Key: ((o.coupon_code IS NOT NULL)), o.order_id
        Sort Method: quicksort  Memory: 1572kB
        Buffers: shared hit=368
        ->  Hash Join  (cost=333.15..928.47 rows=20565 width=15) (actual time=1.268..3.837 rows=20557
loops=1)
              Hash Cond: (oi.order_id = o.order_id)
              Buffers: shared hit=360
              ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.002..0.785 rows=26716 loops=1)
                    Buffers: shared hit=258
              ->  Hash  (cost=237.99..237.99 rows=7613 width=15) (actual time=1.253..1.254 rows=7613
loops=1)
                    Buckets: 8192  Batches: 1  Memory Usage: 376kB
                    Buffers: shared hit=102
                    ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=15) (actual
time=0.003..0.878 rows=7613 loops=1)
                          Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                          Rows Removed by Filter: 2277
                          Buffers: shared hit=102
Planning:
  Buffers: shared hit=276
Planning Time: 0.616 ms
Execution Time: 8.224 ms
```

튜닝 포인트

변경 주문별 금액 선집계 후 쿠폰 그룹 AVG

튜닝 후

```
WITH order_totals AS (
            SELECT
                o.order_id,
                o.coupon_code IS NOT NULL AS used_coupon,
                sum(oi.line_total) AS amount
            FROM ecom.orders o
            JOIN ecom.order_items oi
              ON oi.order_id = o.order_id
            WHERE o.order_status IN ('paid', 'shipped', 'delivered')
            GROUP BY
                o.order_id,
                o.coupon_code IS NOT NULL
        )

        SELECT
            used_coupon,
            count(*) AS order_count,
            avg(amount) AS avg_order_amount
        FROM order_totals
        GROUP BY used_coupon
        ORDER BY used_coupon
```

튜닝 후 실행 결과

Q09 튜닝 후 결과

```
used_coupon | order_count | avg_order_amount
------------+-------------+-----------------
False       | 5796        | 628.88
True        | 1817        | 1,839.69
rows=2 | median execution=5.462 ms
```

튜닝 후 성능 원문 로그 · Q09_plan_after.png (1/2)

```
Q09 쿠폰 사용 여부별 평균 주문 금액 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH order_totals AS (
        SELECT
            o.order_id,
            o.coupon_code IS NOT NULL AS used_coupon,
            sum(oi.line_total) AS amount
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY
            o.order_id,
            o.coupon_code IS NOT NULL
    )

    SELECT
        used_coupon,
        count(*) AS order_count,
        avg(amount) AS avg_order_amount
    FROM order_totals
    GROUP BY used_coupon
    ORDER BY used_coupon

PERFORMANCE RESULT
Sort  (cost=1539.53..1539.53 rows=2 width=41) (actual time=8.816..8.819 rows=2 loops=1)
  Sort Key: ((o.coupon_code IS NOT NULL))
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=365
  ->  HashAggregate  (cost=1539.49..1539.52 rows=2 width=41) (actual time=8.803..8.804 rows=2 loops=1)
        Group Key: ((o.coupon_code IS NOT NULL))
        Batches: 1  Memory Usage: 24kB
        Buffers: shared hit=360
        ->  HashAggregate  (cost=1082.71..1273.04 rows=15226 width=41) (actual time=7.464..8.391
rows=7613 loops=1)
              Group Key: o.order_id, (o.coupon_code IS NOT NULL)
              Batches: 1  Memory Usage: 3601kB
              Buffers: shared hit=360
              ->  Hash Join  (cost=333.15..928.47 rows=20565 width=15) (actual time=1.657..5.164
rows=20557 loops=1)
                    Hash Cond: (oi.order_id = o.order_id)
                    Buffers: shared hit=360
                    ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.003..1.023 rows=26716 loops=1)
                          Buffers: shared hit=258
                    ->  Hash  (cost=237.99..237.99 rows=7613 width=15) (actual time=1.643..1.643
rows=7613 loops=1)
                          Buckets: 8192  Batches: 1  Memory Usage: 374kB
                          Buffers: shared hit=102
                          ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=15) (actual
time=0.004..1.148 rows=7613 loops=1)
                                Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                Rows Removed by Filter: 2277
                                Buffers: shared hit=102
Planning:
  Buffers: shared hit=279
Planning Time: 0.415 ms
Execution Time: 8.916 ms
```

튜닝 후 성능 원문 로그 · Q09_plan_after.png (2/2)

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	8.061 ms	5.462 ms
Planning Time	0.053 ms	0.047 ms
Actual Rows	2	2
Shared Hit	360	360
Shared Read	0	0
Top Plan Node	Aggregate	Sort
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

AOV의 업무 단위가 주문임을 SQL 구조에 직접 반영했다.

마케팅·재무 전달안

쿠폰 사용 주문 AOV는 1,839.69, 미사용 주문은 628.88로 표본상 192.5% 높다. 인과효과가 아니라 관찰 비교다.

Q10. 상위 1% 고객의 최근 60일 매출

요청 부서	CRM·VIP
요청 목적	상위 1% 고객의 최근 60일 매출
판정 기준	누적매출 Top 30 후 60일 집계

튜닝 전

Top 30 고객마다 최근 60일 매출 상관 조회

```
WITH lifetime AS (
            SELECT
                o.customer_id,
                sum(oi.line_total) AS lifetime_revenue
            FROM ecom.orders o
            JOIN ecom.order_items oi
              ON oi.order_id = o.order_id
            WHERE o.order_status IN ('paid', 'shipped', 'delivered')
            GROUP BY o.customer_id
        ),
        top_customers AS (
            SELECT
                customer_id,
                lifetime_revenue
            FROM lifetime
            ORDER BY
                lifetime_revenue DESC,
                customer_id
            LIMIT 30
        )

        SELECT
            tc.customer_id,
            tc.lifetime_revenue,
            (
                SELECT sum(oi.line_total)
                FROM ecom.orders o
                JOIN ecom.order_items oi
                  ON oi.order_id = o.order_id
                WHERE o.customer_id = tc.customer_id
                  AND o.order_status IN ('paid', 'shipped', 'delivered')
                  AND o.order_ts >= now() - interval '60 days'
            ) AS recent_60d_revenue
        FROM top_customers tc
        ORDER BY
            tc.lifetime_revenue DESC,
            tc.customer_id
```

튜닝 전 실행 결과

Q10 튜닝 전 결과

```
customer_id | lifetime_revenue | recent_60d_revenue
------------+------------------+-------------------
19          | 37,803.69        | 37,803.69
7           | 36,318.34        | 36,318.34
3           | 33,809.53        | 30,444.71
25          | 33,689.05        | 32,619.33
1           | 33,226.61        | 30,255.84
10          | 33,080.74        | 32,851.70
21          | 32,305.26        | 31,585.52
15          | 31,575.76        | 30,963.58
... 22 rows omitted
rows=30 | median execution=5.036 ms
```

튜닝 전 성능 원문 로그 · Q10_plan_before.png (1/2)

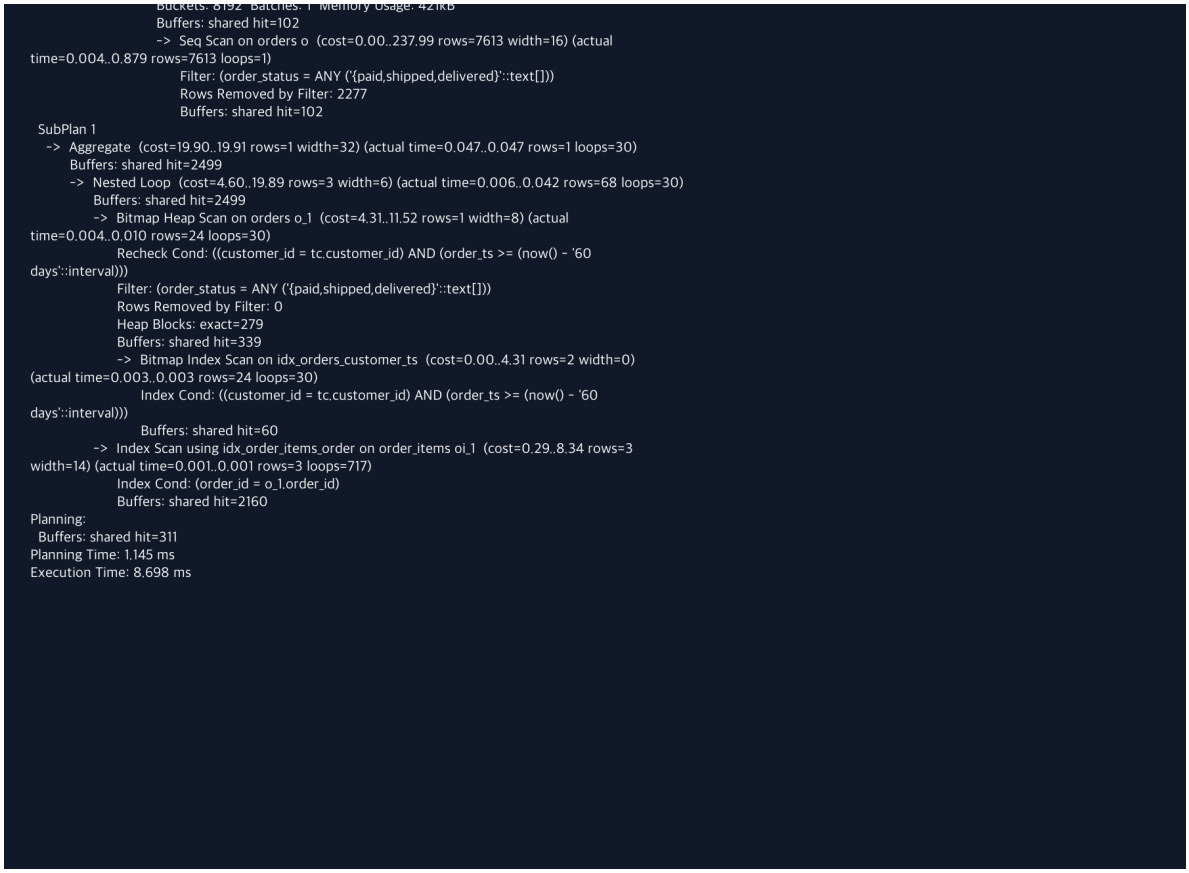
```
Q10 상위 1% 고객의 최근 60일 매출 [before]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH lifetime AS (
        SELECT
          o.customer_id,
          sum(oi.line_total) AS lifetime_revenue
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY o.customer_id
      ),
      top_customers AS (
        SELECT
          customer_id,
          lifetime_revenue
        FROM lifetime
        ORDER BY
          lifetime_revenue DESC,
          customer_id
        LIMIT 30
      )

      SELECT
        tc.customer_id,
        tc.lifetime_revenue,
        (
          SELECT sum(oi.line_total)
          FROM ecom.orders o
          JOIN ecom.order_items oi
            ON oi.order_id = o.order_id
          WHERE o.customer_id = tc.customer_id
            AND o.order_status IN ('paid', 'shipped', 'delivered')
            AND o.order_ts >= now() - interval '60 days'
        ) AS recent_60d_revenue
      FROM top_customers tc
      ORDER BY
        tc.lifetime_revenue DESC,
        tc.customer_id

PERFORMANCE RESULT
Subquery Scan on tc  (cost=1156.39..1753.69 rows=30 width=72) (actual time=7.278..8.608 rows=30 loops=1)
  Buffers: shared hit=2865
  ->  Limit  (cost=1156.39..1156.47 rows=30 width=40) (actual time=7.186..7.190 rows=30 loops=1)
        Buffers: shared hit=366
        ->  Sort  (cost=1156.39..1163.83 rows=2976 width=40) (actual time=7.185..7.187 rows=30 loops=1)
              Sort Key: (sum(oi.line_total)) DESC, o.customer_id
              Sort Method: top-N heapsort  Memory: 27kB
              Buffers: shared hit=366
              ->  HashAggregate  (cost=1031.30..1068.50 rows=2976 width=40) (actual time=6.552..6.902
rows=2250 loops=1)
                    Group Key: o.customer_id
                    Batches: 1  Memory Usage: 1137kB
                    Buffers: shared hit=360
                    ->  Hash Join  (cost=333.15..928.47 rows=20565 width=14) (actual time=1.284..4.740
rows=20557 loops=1)
                          Hash Cond: (oi.order_id = o.order_id)
                          Buffers: shared hit=360
                          ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.004..1.039 rows=26716 loops=1)
                                Buffers: shared hit=258
                          ->  Hash  (cost=237.99..237.99 rows=7613 width=16) (actual time=1.265..1.265
rows=7613 loops=1)
                                Buckets: 8192  Batches: 1  Memory Usage: 421kB
```

튜닝 전 성능 원문 로그 · Q10_plan_before.png (2/2)



```
                        Buckets: 8192  Batches: 1  Memory Usage: 421kB
                        Buffers: shared hit=102
                        ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=16) (actual
time=0.004..0.879 rows=7613 loops=1)
                              Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                              Rows Removed by Filter: 2277
                              Buffers: shared hit=102
  SubPlan 1
    ->  Aggregate  (cost=19.90..19.91 rows=1 width=32) (actual time=0.047..0.047 rows=1 loops=30)
          Buffers: shared hit=2499
          ->  Nested Loop  (cost=4.60..19.89 rows=3 width=6) (actual time=0.006..0.042 rows=68 loops=30)
                Buffers: shared hit=2499
                ->  Bitmap Heap Scan on orders o_1  (cost=4.31..11.52 rows=1 width=8) (actual
time=0.004..0.010 rows=24 loops=30)
                      Recheck Cond: ((customer_id = tc.customer_id) AND (order_ts >= (now() - '60
days'::interval)))
                      Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                      Rows Removed by Filter: 0
                      Heap Blocks: exact=279
                      Buffers: shared hit=339
                      ->  Bitmap Index Scan on idx_orders_customer_ts  (cost=0.00..4.31 rows=2 width=0)
(actual time=0.003..0.003 rows=24 loops=30)
                            Index Cond: ((customer_id = tc.customer_id) AND (order_ts >= (now() - '60
days'::interval)))
                            Buffers: shared hit=60
                ->  Index Scan using idx_order_items_order on order_items oi_1  (cost=0.29..8.34 rows=3
width=14) (actual time=0.001..0.001 rows=3 loops=717)
                      Index Cond: (order_id = o_1.order_id)
                      Buffers: shared hit=2160
Planning:
  Buffers: shared hit=311
Planning Time: 1.145 ms
Execution Time: 8.698 ms
```

튜닝 포인트

변경 한 번의 고객 집계에서 누적·60일 매출을 조건부 합산

튜닝 후

```
WITH customer_sales AS (
            SELECT
                o.customer_id,
                sum(oi.line_total) AS lifetime_revenue,
                sum(oi.line_total) FILTER (
                    WHERE o.order_ts >= now() - interval '60 days'
                ) AS recent_60d_revenue
            FROM ecom.orders o
            JOIN ecom.order_items oi
              ON oi.order_id = o.order_id
            WHERE o.order_status IN ('paid', 'shipped', 'delivered')
            GROUP BY o.customer_id
        )

        SELECT
            customer_id,
            lifetime_revenue,
            recent_60d_revenue
        FROM customer_sales
        ORDER BY
            lifetime_revenue DESC,
            customer_id
        LIMIT 30
```

튜닝 후 실행 결과

Q10 튜닝 후 결과

```
customer_id | lifetime_revenue | recent_60d_revenue
------------+------------------+-------------------
19          | 37,803.69        | 37,803.69
7           | 36,318.34        | 36,318.34
3           | 33,809.53        | 30,444.71
25          | 33,689.05        | 32,619.33
1           | 33,226.61        | 30,255.84
10          | 33,080.74        | 32,851.70
21          | 32,305.26        | 31,585.52
15          | 31,575.76        | 30,963.58
... 22 rows omitted
rows=30 | median execution=6.580 ms
```

튜닝 후 성능 원문 로그 · Q10_plan_after.png (1/2)

```
Q10 상위 1% 고객의 최근 60일 매출 [after]

PERFORMANCE SQL
EXPLAIN (ANALYZE, BUFFERS)
WITH customer_sales AS (
        SELECT
            o.customer_id,
            sum(oi.line_total) AS lifetime_revenue,
            sum(oi.line_total) FILTER (
                WHERE o.order_ts >= now() - interval '60 days'
            ) AS recent_60d_revenue
        FROM ecom.orders o
        JOIN ecom.order_items oi
          ON oi.order_id = o.order_id
        WHERE o.order_status IN ('paid', 'shipped', 'delivered')
        GROUP BY o.customer_id
    )

    SELECT
        customer_id,
        lifetime_revenue,
        recent_60d_revenue
    FROM customer_sales
    ORDER BY
        lifetime_revenue DESC,
        customer_id
    LIMIT 30

PERFORMANCE RESULT
Limit  (cost=1369.48..1369.56 rows=30 width=72) (actual time=12.920..12.928 rows=30 loops=1)
  Buffers: shared hit=366
  ->  Sort  (cost=1369.48..1376.92 rows=2976 width=72) (actual time=12.919..12.921 rows=30 loops=1)
        Sort Key: (sum(oi.line_total)) DESC, o.customer_id
        Sort Method: top-N heapsort  Memory: 28kB
        Buffers: shared hit=366
        ->  HashAggregate  (cost=1236.95..1281.59 rows=2976 width=72) (actual time=12.113..12.627
rows=2250 loops=1)
              Group Key: o.customer_id
              Batches: 1  Memory Usage: 1649kB
              Buffers: shared hit=360
              ->  Hash Join  (cost=333.15..928.47 rows=20565 width=22) (actual time=2.294..6.494
rows=20557 loops=1)
                    Hash Cond: (oi.order_id = o.order_id)
                    Buffers: shared hit=360
                    ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual
time=0.006..1.208 rows=26716 loops=1)
                          Buffers: shared hit=258
                    ->  Hash  (cost=237.99..237.99 rows=7613 width=24) (actual time=2.272..2.272
rows=7613 loops=1)
                          Buckets: 8192  Batches: 1  Memory Usage: 481kB
                          Buffers: shared hit=102
                          ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=24) (actual
time=0.004..1.527 rows=7613 loops=1)
                                Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                Rows Removed by Filter: 2277
                                Buffers: shared hit=102
Planning:
  Buffers: shared hit=289
Planning Time: 0.833 ms
Execution Time: 12.991 ms
```

튜닝 후 성능 원문 로그 · Q10_plan_after.png (2/2)

튜닝 전후 실행계획 비교

측정 항목	튜닝 전	튜닝 후
Execution Time	5.036 ms	6.580 ms
Planning Time	0.088 ms	0.043 ms
Actual Rows	30	30
Shared Hit	2,856	360
Shared Read	0	0
Top Plan Node	Subquery Scan	Limit
결과 동일	-	**YES**

업무 판단 및 요청 부서 전달안

튜닝 판단

고객당 반복 스캔을 한 번의 집계로 줄였다.

CRM·VIP 전달안

상위 1%는 30명으로 고정했다. 1위 고객 ID 19의 누적매출은 37,803.69, 최근 60일 매출은 37,803.69이며 Top 30의 최근 매출 합계는 820,198.65다.

Q11. 안전한 나눗셈 함수 비교

검증 목적	분모 0·NULL 처리 차이 확인
비교 함수	ecom.f_safe_div · ecom.safe_div
적용 판단	평균 계산은 관측 없음과 실제 0을 구분

검증 SQL 및 결과

```
SELECT
            ecom.f_safe_div(10, 2) AS f_normal,
            ecom.f_safe_div(10, 0) AS f_zero,
            ecom.safe_div(10, 2) AS safe_normal,
            ecom.safe_div(10, 0) AS safe_zero,
            ecom.safe_div(10, NULL) AS safe_null
```

Q11 안전한 나눗셈 함수 비교

SQL
SELECT
ecom.f_safe_div(10, 2) AS f_normal,
ecom.f_safe_div(10, 0) AS f_zero,
ecom.safe_div(10, 2) AS safe_normal,
ecom.safe_div(10, 0) AS safe_zero,
ecom.safe_div(10, NULL) AS safe_null

RESULT

f_normal	f_zero	safe_normal	safe_zero	safe_null
5.0000000000000000	0	5.0000000000000000	NULL	NULL

total rows: 1

결과 정상 분모에서는 두 함수 모두 5를 반환했다. 분모가 0이면 f_safe_div는 0, safe_div는 NULL을 반환하며 safe_div는 NULL 분모도 NULL로 유지한다.

선택 평균값에서 '관측 없음'과 실제 0을 구분하기 위해 Q2·Q6에는 ecom.safe_div를 사용했다.

부록 A. 조인 방식 비교

같은 고객별 매출 집계 쿼리에서 PostgreSQL의 조인 옵션을 각각 강제해 실제 실행계획을 비교했다. 목적은 특정 조인을 암기하는 것이 아니라, 입력 행 수·선택도·인덱스·반복 탐색 비용을 보고 옵티마이저의 선택을 검증하는 것이다. 수치는 현재 시드 데이터와 캐시 상태에서의 1회 관측값이다.

이 쿼리에서 조인 전략을 고민한 이유

- orders 9,890건 중 유효 상태 7,613건이 남아 외부 입력이 작지 않고, order_items 26,716건과 1:N으로 결합된다.
- 조인 조건은 order_id 동등 비교이므로 Hash Join의 기본 조건에 맞지만, 양쪽에 order_id 인덱스가 있어 Merge Join과 Nested Loop도 후보가 된다.
- 최종 집계는 2,250명이지만 조인 단계에서는 20,557행을 처리하므로, 출력 행 수만 보고 Nested Loop가 적합하다고 판단하면 안 된다.

확인 절차와 판독 포인트

1. 비교 통제	동일 SQL·동일 데이터에서 SET LOCAL로 하나의 조인 방식만 활성화
2. 결과 동일성	세 계획 모두 조인 20,557행, 최종 2,250행인지 확인
3. 추정 품질	예상 20,565행 대비 실제 20,557행으로 행 수 오차가 매우 작음
4. 실제 비용	Execution Time·Buffers hit·loops·스캔 방식·Hash Batch/메모리를 함께 비교
5. 해석 제한	조인 강제는 원인 확인용이며, 운영 SQL에서는 통계정보와 옵티마이저 선택을 우선

방식	실행시간	Buffers hit	적합한 상황	이번 결과
Nested Loop	19.833 ms	23,069	외부 결과가 작고 내부 인덱스가 유효	orders 7,613행마다 order_items 인덱스 탐색
Hash Join	12.881 ms	360	큰 동등 조인·정렬 불필요	현재 데이터에서 가장 빠르고 버퍼 접근 최소
Merge Join	13.837 ms	442	양쪽 입력이 조인 키로 정렬	PK와 order_id 인덱스 순서를 활용

실측 결과 해석

Nested Loop orders 7,613행마다 idx_order_items_order를 탐색해 내부 Index Scan loops=7,613이 발생했다. 건당 탐색은 빨랐지만 누적 Buffers hit=23,069, 19.833ms로 가장 비싼 계획이었다.

Hash Join 필터를 통과한 orders 7,613행으로 421kB 해시를 만들고 order_items를 1회 순차 스캔했다. Batches=1이라 디스크 spill이 없었고, Buffers hit=360, 12.881ms로 가장 효율적이었다.

Merge Join orders_pkey와 idx_order_items_order의 정렬 순서를 활용해 별도 Sort는 없었다. Buffers hit=442, 13.837ms로 Hash Join에 근접했지만 양쪽 Index Scan 비용 때문에 약간 느렸다.

결론 현 분포에서는 Hash Join이 Nested Loop보다 6.952ms(35.1%) 빨랐고 버퍼 hit를 98.4% 줄였다. 다만 기간 필터 등으로 외부 행이 소수로 줄면 Nested Loop를, 양쪽이 이미 조인 키 순으로 정렬된 대량 입력이면 Merge Join을 다시 검토한다.

Bitmap Heap Scan은 어떻게 볼 것인가

Bitmap Heap Scan은 Hash/Nested Loop/Merge와 같은 조인 알고리즘이 아니라 테이블 접근 방식이다. 인덱스 조건에 맞는 행이 여러 페이지에 퍼져 있을 때 위치를 비트맵으로 모아 heap 페이지를 묶어 읽는다. 이번 비교에서는 orders 상태 필터가 많은 행을 남겨 Seq Scan이 선택됐고 Bitmap Heap Scan은 나타나지 않았다. 기간·상태 조건의 선택도가 중간 수준으로 낮아지면 Bitmap Index Scan과 함께 선택되는지 확인할 가치가 있다.

실행계획 증적

Nested Loop: 내부 Index Scan loops=7,613과 Buffers hit=23,069 확인

```
nested_loop

SQL
SELECT o.customer_id, count(*) AS item_count, sum(oi.line_total) AS revenue
    FROM ecom.orders o JOIN ecom.order_items oi USING (order_id)
    WHERE o.order_status IN ('paid','shipped','delivered')
    GROUP BY o.customer_id

PLAN
HashAggregate  (cost=4461.04..4498.24 rows=2976 width=48) (actual time=19.210..19.721 rows=2250 loops=1)
  Group Key: o.customer_id
  Batches: 1  Memory Usage: 1137kB
  Buffers: shared hit=23069
  ->  Nested Loop  (cost=0.29..4306.80 rows=20565 width=14) (actual time=0.013..14.662 rows=20557
loops=1)
        Buffers: shared hit=23069
        ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=16) (actual time=0.005..2.094
rows=7613 loops=1)
              Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
              Rows Removed by Filter: 2277
              Buffers: shared hit=102
        ->  Index Scan using idx_order_items_order on order_items oi  (cost=0.29..0.50 rows=3 width=14)
(actual time=0.001..0.001 rows=3 loops=7613)
              Index Cond: (order_id = o.order_id)
              Buffers: shared hit=22967
Planning:
  Buffers: shared hit=252
Planning Time: 0.712 ms
Execution Time: 19.833 ms
```

Hash Join: Hash Batches=1, Memory Usage=421kB, Buffers hit=360 확인

```
hash_join

SQL
SELECT o.customer_id, count(*) AS item_count, sum(oi.line_total) AS revenue
    FROM ecom.orders o JOIN ecom.order_items oi USING (order_id)
    WHERE o.order_status IN ('paid','shipped','delivered')
    GROUP BY o.customer_id

PLAN
HashAggregate  (cost=1082.71..1119.91 rows=2976 width=48) (actual time=12.174..12.769 rows=2250 loops=1)
  Group Key: o.customer_id
  Batches: 1  Memory Usage: 1137kB
  Buffers: shared hit=360
  ->  Hash Join  (cost=333.15..928.47 rows=20565 width=14) (actual time=2.317..8.581 rows=20557 loops=1)
        Hash Cond: (oi.order_id = o.order_id)
        Buffers: shared hit=360
        ->  Seq Scan on order_items oi  (cost=0.00..525.16 rows=26716 width=14) (actual time=0.003..1.608
rows=26716 loops=1)
              Buffers: shared hit=258
        ->  Hash  (cost=237.99..237.99 rows=7613 width=16) (actual time=2.301..2.302 rows=7613 loops=1)
              Buckets: 8192  Batches: 1  Memory Usage: 421kB
              Buffers: shared hit=102
              ->  Seq Scan on orders o  (cost=0.00..237.99 rows=7613 width=16) (actual time=0.004..1.545
rows=7613 loops=1)
                    Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                    Rows Removed by Filter: 2277
                    Buffers: shared hit=102
Planning:
  Buffers: shared hit=252
Planning Time: 0.621 ms
Execution Time: 12.881 ms
```

Merge Join: 양쪽 Index Scan과 별도 Sort 없음 확인

```
merge_join

SQL
SELECT o.customer_id, count(*) AS item_count, sum(oi.line_total) AS revenue
    FROM ecom.orders o JOIN ecom.order_items oi USING (order_id)
    WHERE o.order_status IN ('paid','shipped','delivered')
    GROUP BY o.customer_id

PLAN
HashAggregate  (cost=1734.46..1771.66 rows=2976 width=48) (actual time=13.252..13.718 rows=2250 loops=1)
  Group Key: o.customer_id
  Batches: 1  Memory Usage: 1137kB
  Buffers: shared hit=442
  ->  Merge Join  (cost=0.57..1580.22 rows=20565 width=14) (actual time=0.022..9.031 rows=20557 loops=1)
        Merge Cond: (o.order_id = oi.order_id)
        Buffers: shared hit=442
        ->  Index Scan using orders_pkey on orders o  (cost=0.29..406.72 rows=7613 width=16) (actual
time=0.010..2.165 rows=7613 loops=1)
              Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
              Rows Removed by Filter: 2277
              Buffers: shared hit=130
        ->  Index Scan using idx_order_items_order on order_items oi  (cost=0.29..882.03 rows=26716
width=14) (actual time=0.007..3.042 rows=26716 loops=1)
              Buffers: shared hit=312
Planning:
  Buffers: shared hit=276
Planning Time: 0.732 ms
Execution Time: 13.837 ms
```

비교 SQL

```sql
SELECT
    o.customer_id,
    count(*) AS item_count,
    sum(oi.line_total) AS revenue
FROM ecom.orders o
JOIN ecom.order_items oi
  ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid', 'shipped', 'delivered')
GROUP BY o.customer_id;
```

부록 B. Materialized View

반복 요청이 많은 일별 GMV 집계를 미리 저장해 조회 비용을 줄이는 구조다. 일반 View와 달리 결과를 물리적으로 보관하므로 원본 변경 후 명시적으로 갱신해야 한다.

생성 및 갱신 스크립트

```
CREATE MATERIALIZED VIEW IF NOT EXISTS ecom.mv_daily_gmv AS
SELECT
    date_trunc('day', o.order_ts) AS day,
    sum(oi.line_total) AS gmv
FROM ecom.orders o
JOIN ecom.order_items oi
  ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid', 'shipped', 'delivered')
GROUP BY date_trunc('day', o.order_ts);

CREATE INDEX IF NOT EXISTS idx_mv_daily_gmv_day
ON ecom.mv_daily_gmv(day);

REFRESH MATERIALIZED VIEW ecom.mv_daily_gmv;

SELECT
    count(*) AS days,
    sum(gmv) AS gmv,
    max(day) AS latest_day
FROM ecom.mv_daily_gmv;
```

검증 결과 트랜잭션 안에서 테스트 매출 123.45를 추가하자 원본 GMV만 증가하고 MV는 이전 값을 유지했다. REFRESH 후 두 합계가 같아졌으며, 마지막 ROLLBACK으로 테스트 행이 남지 않았음을 확인했다.

운영 판단 Materialized View는 자동 갱신되지 않는다. 일 배치나 적재 완료 이벤트 뒤에 REFRESH를 실행하고, 조회 중단을

부록 C. 최종 인사이트

랜덤 데이터에서 배운 점

- 숫자 자체보다 정의가 먼저다. 매출 상태·기간·주문 단위를 합의하지 않으면 같은 요청에도 다른 답이 나온다.
- 1:N 조인은 주문 수와 AOV를 쉽게 부풀린다. 주문별 선집계는 성능보다 먼저 정확성을 지키는 장치다.
- 튜닝 후 SQL이 항상 더 빠르지는 않았다. 작은 테이블에서는 Seq Scan이나 CTE 인라인이 합리적이며, 실행계획과 버퍼를 함께 봐야 한다.
- 랜덤 시드는 결과값을 바꾸지만 검증 가능한 로직은 유지된다. 따라서 전달 자료에는 기준시각·조건·단위·가정을 함께 적어야 한다.
- 상관관계와 인과관계를 구분해야 한다. 쿠폰 주문의 높은 AOV는 쿠폰 효과일 수도 있지만 고가 상품 편향이나 시드 생성 규칙의 영향일 수도 있다.

DBMS별 실행계획 확인 도구(선택)

DBMS	대표 도구	확인 포인트
PostgreSQL	EXPLAIN (ANALYZE, BUFFERS)	실제 행·시간·버퍼와 예상치 비교
MySQL	EXPLAIN ANALYZE	실제 반복·행 수와 비용 확인
Oracle	DBMS_XPLAN	실행 통계와 접근 경로 확인
SQL Server	Actual Execution Plan	연산자 비용·실제/예상 행 차이 확인